Differential Equations Workbook Solutions

Higher order equations

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HOMOGENEOUS HIGHER ORDER EQUATIONS

■ 1. Find the general solution to the third order homogeneous differential equation.

$$y''' - 3y'' - y' + 3y = 0$$

Solution:

We'll rewrite the differential equation as its associated characteristic equation, and then use a calculator to factor the polynomial.

$$r^3 - 3r^2 - r + 3 = 0$$

$$(r + 1)(r - 1)(r - 3) = 0$$

From the factored form of the polynomial, we can see that the equation has distinct real roots $r_1 = -1$, $r_2 = 1$, and $r_3 = 3$. And the general solution is therefore

$$y(t) = c_1 e^{-t} + c_2 e^{t} + c_3 e^{3t}$$

■ 2. Find the general solution to the third order homogeneous differential equation.

$$y''' + 6y'' + 8y' = 0$$

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Solution:

We'll rewrite the differential equation as its associated characteristic equation, and then use a calculator to factor the polynomial.

$$r^3 + 6r^2 + 8r = 0$$

$$r(r+2)(r+4) = 0$$

From the factored form of the polynomial, we can see that the equation has distinct real roots $r_1 = 0$, $r_2 = -2$, and $r_3 = -4$. And the general solution is therefore

$$y(t) = c_1 + c_2 e^{-2t} + c_3 e^{-4t}$$

■ 3. Find the general solution to the third order homogeneous differential equation.

$$y''' + 15y'' + 75y' + 125y = 0$$

Solution:

We'll rewrite the differential equation as its associated characteristic equation, and then use a calculator to factor the polynomial.

$$r^3 + 15r^2 + 75r + 125 = 0$$

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$$(r+5)^3 = 0$$

From the factored form of the polynomial, we can see that the equation has three equal real roots $r_1 = r_2 = r_3 = -5$. And the general solution is therefore

$$y(t) = c_1 e^{-5t} + c_2 t e^{-5t} + c_3 t^2 e^{-5t}$$

■ 4. Find the general solution to the third order homogeneous differential equation.

$$y''' - 5y'' + 12y' - 8y = 0$$

Solution:

We'll rewrite the differential equation as its associated characteristic equation, and then use a calculator to factor the polynomial.

$$r^3 - 5r^2 + 12r - 8 = 0$$

$$(r-1)(r^2 - 4r + 8) = 0$$

From the factored form of the polynomial, we can see that the equation has one real root $r_1 = 1$, and two complex conjugate roots $r_{2,3} = 2 \pm 2i$. And the general solution is therefore

$$y(t) = c_1 e^t + c_2 e^{2t} \cos(2t) + c_3 e^{2t} \sin(2t)$$

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■ 5. Find the general solution to the fifth order homogeneous differential equation.

$$y^{(5)} - 2y^{(4)} + 2y''' = 0$$

Solution:

We'll rewrite the differential equation as its associated characteristic equation, and then use a calculator to factor the polynomial.

$$r^5 - 2r^4 + 2r^3 = 0$$

$$r^3(r^2 - 2r + 2) = 0$$

From the factored form of the polynomial, we can see that the equation has three equal real roots $r_1 = r_2 = r_3 = 0$, and two complex conjugate roots $r_{4,5} = 1 \pm i$. And the general solution is therefore

$$y(t) = c_1 + c_2 t + c_3 t^2 + c_4 e^t \cos t + c_5 e^t \sin t$$

■ 6. Find the general solution to the seventh order homogeneous differential equation.

$$y^{(7)} + 15y^{(6)} + 125y^{(5)} + 595y^{(4)} + 1{,}795y''' + 2{,}861y'' + 2{,}175y' + 625y = 0$$

Solution:

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We'll rewrite the differential equation as its associated characteristic equation, and then use a calculator to factor the polynomial.

$$r^7 + 15r^6 + 125r^5 + 595r^4 + 1{,}795r^3 + 2{,}861r^2 + 2{,}175r + 625 = 0$$

$$(r+1)^3(r^2 + 6r + 25)^2 = 0$$

From the factored form of the polynomial, we can see that the equation has three equal real roots $r_1 = r_2 = r_3 = -1$, and a pair of complex conjugate roots with multiplicity two, $r_{4,5} = -3 \pm 4i$ and $r_{6,7} = -3 \pm 4i$. And the general solution is therefore

$$y(t) = c_1 e^{-t} + c_2 t e^{-t} + c_3 t^2 e^{-t} + c_4 e^{-3t}\cos(4t) + c_5 e^{-3t}\sin(4t)$$

$$+ c_6 t e^{-3t}\cos(4t) + c_7 t e^{-3t}\sin(4t)$$

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UNDETERMINED COEFFICIENTS FOR HIGHER ORDER EQUATIONS

■ 1. Find the general solution to the third order differential equation.

$$y''' + y'' - 4y' - 4y = 8t^2$$

Solution:

Find the roots of the characteristic equation.

$$r^3 + r^2 - 4r - 4 = 0$$

$$(r+1)(r^2 - 4) = 0$$

$$(r+1)(r+2)(r-2) = 0$$

$$r = -1,\ 2,\ -2$$

With three distinct roots, the complementary solution is

$$y_c(t) = c_1 e^{-t} + c_2 e^{2t} + c_3 e^{-2t}$$

Our guess for the particular solution and its derivatives will be

$$y_p(t) = At^2 + Bt + C$$

$$y_p'(t) = 2At + B$$

$$y_p''(t) = 2A$$

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$$y_p'''(t) = 0$$

Plugging these into the differential equation gives

$$0 + 2A - 8At - 4B - 4At^2 - 4Bt - 4C = 8t^2$$

$$-4At^2 - (8A + 4B)t + 2A - 4B - 4C = 8t^2$$

Equating coefficients gives us the system of equations

$$-4A = 8$$

$$-8A - 4B = 0$$

$$2A - 4B - 4C = 0$$

From this system, we find $A = -2$, $B = 4$, and $C = -5$, so the particular solution will be

$$y_p(t) = -2t^2 + 4t - 5$$

The general solution will be the sum of the complementary and particular solutions.

$$y(t) = y_c(t) + y_p(t)$$

$$y(t) = c_1e^{-t} + c_2e^{2t} + c_3e^{-2t} - 2t^2 + 4t - 5$$

■ 2. Find the general solution to the third order differential equation.

$$y''' - 6y'' + 9y' - 4y = 1 + e^{2t} + 9e^{4t}$$

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Solution:

Find the roots of the characteristic equation.

$$r^3 - 6r^2 + 9r - 4 = 0$$

$$(r-1)^2(r-4) = 0$$

$$r = 1,\ 1,\ 4$$

With a pair of repeated roots and one distinct root, the complementary solution is

$$y_c(t) = c_1e^t + c_2te^t + c_3e^{4t}$$

Our guess for the particular solution would be

$$y_p = A + Be^{2t} + Ce^{4t}$$

but Ce^{4t} overlaps with c_3e^{4t} from the complementary solution. So we multiply that term by t to get Cte^{4t}. So the particular solution and its derivatives will be

$$y_p(t) = A + Be^{2t} + Cte^{4t}$$

$$y_p'(t) = 2Be^{2t} + Ce^{4t} + 4Cte^{4t}$$

$$y_p''(t) = 4Be^{2t} + 8Ce^{4t} + 16Cte^{4t}$$

$$y_p'''(t) = 8Be^{2t} + 48Ce^{4t} + 64Cte^{4t}$$

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Plugging these into the differential equation gives

$$8Be^{2t} + 64Cte^{4t} + 48Ce^{4t} - 24Be^{2t} - 48Ce^{4t} - 96Cte^{4t}$$

$$+18Be^{2t} + 9Ce^{4t} + 36Cte^{4t} - 4A - 4Be^{2t} - 4Cte^{4t}$$

$$= 1 + e^{2t} + 9e^{4t}$$

$$-2Be^{2t} + 9Ce^{4t} - 4A = 1 + e^{2t} + 9e^{4t}$$

Equating coefficients gives us the system of equations

$$-4A = 1$$

$$-2B = 1$$

$$9C = 9$$

From this system, we find $A = -1/4$, $B = -1/2$, and $C = 1$, so the particular solution will be

$$y_p(t) = -\frac{1}{4} - \frac{1}{2}e^{2t} + te^{4t}$$

The general solution will be the sum of the complementary and particular solutions.

$$y(t) = y_c(t) + y_p(t)$$

$$y(t) = c_1e^t + c_2te^t + c_3e^{4t} - \frac{1}{4} - \frac{1}{2}e^{2t} + te^{4t}$$

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■ 3. Find the general solution to the differential equation. Hint: $x^4 + 4x^3 + 7x^2 + 6x + 2 = (x + 1)^2(x^2 + 2x + 2)$.

$$y'''' + 4y''' + 7y'' + 6y' + 2y = e^{-t}(\cos t + 2\sin t + 4)$$

Solution:

Find the roots of the characteristic equation.

$$r^4 + 4r^3 + 7r^2 + 6r + 2 = 0$$

$$(r + 1)^2(r^2 + 2r + 2) = 0$$

$$(r + 1)^2(r + i + 1)(r - i + 1) = 0$$

$$r = -1,\ -1,\ -1 \pm i$$

With a pair of repeated roots and a pair of complex conjugate roots, the complementary solution is

$$y_c(t) = c_1e^{-t} + c_2te^{-t} + c_3e^{-t}\cos t + c_4e^{-t}\sin t$$

Our guess for the particular solution would be

$$y_p = Ae^{-t}\cos t + Be^{-t}\sin t + Ce^{-t}$$

but $Ae^{-t}\cos t$ and $Be^{-t}\sin t$ overlap with $c_3e^{-t}\cos t$ and $c_4e^{-t}\sin t$ from the complementary solution, so we multiply those terms by t. And Ce^{-t} overlaps with c_1e^{-t}, so we'll multiply it by t^2 (if we only multiplied by t, then we'd still have an overlap with c_2te^{-t}). So the particular solution and its derivatives will be

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$$y_p(t) = Ate^{-t}\cos t + Bte^{-t}\sin t + Ct^2e^{-t}$$

$$y_p'(t) = e^{-t}[\cos t(A - At + Bt) + \sin t(B - Bt - At) + 2Ct - Ct^2]$$

$$y_p''(t) = e^{-t}[(2At - 2A - 2B)\sin t + (-2Bt - 2A + 2B)\cos t + C(t^2 - 4t + 2)]$$

$$y_p'''(t) = e^{-t}[(-2At + 2Bt + 6A)\sin t + (2At + 2Bt - 6B)\cos t + C(-t^2 + 6t - 6)]$$

$$y_p''''(t) = e^{-t}[(-4Bt - 8A + 8B)\sin t + (-4At + 8A + 8B)\cos t + C(t^2 - 8t + 12)]$$

Plugging these into the differential equation gives

$$e^{-t}[(-4Bt - 8A + 8B)\sin t + (-4At + 8A + 8B)\cos t + C(t^2 - 8t + 12)]$$

$$+4e^{-t}[(-2At + 2Bt + 6A)\sin t + (2At + 2Bt - 6B)\cos t + C(-t^2 + 6t - 6)]$$

$$+7e^{-t}[(2At - 2A - 2B)\sin t + (-2Bt - 2A + 2B)\cos t + C(t^2 - 4t + 2)]$$

$$+6e^{-t}[\cos t(A - At + Bt) + \sin t(B - Bt - At) + 2Ct - Ct^2]$$

$$+2[Ate^{-t}\cos t + Bte^{-t}\sin t + Ct^2e^{-t}]$$

$$= e^{-t}(\cos t + 2\sin t + 4)$$

$$e^{-t}[(-4Bt - 8A + 8B)\sin t + (-4At + 8A + 8B)\cos t + C(t^2 - 8t + 12)]$$

$$+e^{-t}[(-8At + 8Bt + 24A)\sin t + (8At + 8Bt - 24B)\cos t + C(-4t^2 + 24t - 24)]$$

$$+e^{-t}[(14At - 14A - 14B)\sin t + (-14Bt - 14A + 14B)\cos t + C(7t^2 - 28t + 14)]$$

$$+e^{-t}[(6A - 6At + 6Bt)\cos t + (6B - 6Bt - 6At)\sin t + C(12t - 6t^2)]$$

$$+2Ate^{-t}\cos t + 2Bte^{-t}\sin t + 2Ct^2e^{-t}$$

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$$= e^{-t}(\cos t + 2\sin t + 4)$$

$$e^{-t}[(-4Bt - 8A + 8B - 8At + 8Bt + 24A + 14At - 14A - 14B + 6B - 6Bt - 6At + 2Bt)\sin t$$

$$+(-4At + 8A + 8B + 8At + 8Bt - 24B - 14Bt - 14A + 14B + 6A - 6At + 6Bt + 2At)\cos t$$

$$+C(t^2 - 8t + 12 - 4t^2 + 24t - 24 + 7t^2 - 28t + 14 + 12t - 6t^2 + 2t^2)]$$

$$= e^{-t}(\cos t + 2\sin t + 4)$$

$$e^{-t}(2A\sin t - 2B\cos t + 2C) = e^{-t}(\cos t + 2\sin t + 4)$$

Equating coefficients gives us the system of equations

$$2A = 2$$

$$-2B = 1$$

$$2C = 4$$

From this system, we find $A = 1$, $B = -1/2$, and $C = 2$, so the particular solution will be

$$y_p(t) = te^{-t}\cos t - \frac{1}{2}te^{-t}\sin t + 2t^2e^{-t}$$

The general solution will be the sum of the complementary and particular solutions.

$$y(t) = y_c(t) + y_p(t)$$

$$y(t) = c_1e^{-t} + c_2te^{-t} + c_3e^{-t}\cos t + c_4e^{-t}\sin t$$

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$$+te^{-t}\cos t - \frac{1}{2}te^{-t}\sin t + 2t^2e^{-t}$$

■ 4. Find the general solution to the differential equation.

$$y''' + 9y'' + 27y' + 27y = 20t^2e^{-3t}$$

Solution:

Find the roots of the characteristic equation.

$$r^3 + 9r^2 + 27r + 27 = 0$$

$$(r+3)^3 = 0$$

$$r = -3,\ -3,\ -3$$

With a repeated root with multiplicity three, the complementary solution is

$$y_c(t) = c_1e^{-3t} + c_2te^{-3t} + c_3t^2e^{-3t}$$

Our guess for the particular solution would be

$$y_p(t) = (At^2 + Bt + C)e^{-3t}$$

$$y_p(t) = At^2e^{-3t} + Bte^{-3t} + Ce^{-3t}$$

but all three of the terms in this guess overlap with a term in the complementary solution. Multiplying the particular solution by t would eliminate the overlap in one term, multiplying by t^2 would eliminate the

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overlap in two terms, but multiplying by t^3 is the only way to eliminate the overlap in all three terms. So the particular solution and its derivatives will be

$$y_p(t) = (At^5 + Bt^4 + Ct^3)e^{-3t}$$

$$y_p'(t) = (-3At^5 + (5A - 3B)t^4 + (4B - 3C)t^3 + 3Ct^2)e^{-3t}$$

$$y_p''(t) = (9At^5 + (-30A + 9B)t^4 + (20A - 24B + 9C)t^3 + (12B - 18C)t^2 + 6Ct)e^{-3t}$$

$$y_p'''(t) = [-27At^5 + (135A - 27B)t^4 + (-180A + 108B - 27C)t^3$$

$$+(60A - 108B + 81C)t^2 + (24B - 54C)t + 6C]e^{-3t}$$

Plugging these into the differential equation gives

$$[-27At^5 + (135A - 27B)t^4 + (-180A + 108B - 27C)t^3$$

$$+(60A - 108B + 81C)t^2 + (24B - 54C)t + 6C]e^{-3t}$$

$$+9(9At^5 + (-30A + 9B)t^4 + (20A - 24B + 9C)t^3 + (12B - 18C)t^2 + 6Ct)e^{-3t}$$

$$+27(-3At^5 + (5A - 3B)t^4 + (4B - 3C)t^3 + 3Ct^2)e^{-3t}$$

$$+27(At^5 + Bt^4 + Ct^3)e^{-3t} = 20t^2e^{-3t}$$

$$(60At^2 + 24Bt + 6C)e^{-3t} = 20t^2e^{-3t}$$

Equating coefficients gives us the system of equations

$$60A = 20$$

$$24B = 0$$

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$$6C = 0$$

From this system, we find $A = 1/3$, $B = 0$, and $C = 0$, so the particular solution will be

$$y_p(t) = \frac{1}{3}t^5e^{-3t}$$

The general solution will be the sum of the complementary and particular solutions.

$$y(t) = y_c(t) + y_p(t)$$

$$y(t) = c_1e^{-3t} + c_2te^{-3t} + c_3t^2e^{-3t} + \frac{1}{3}t^5e^{-3t}$$

5. Find the general solution to the differential equation.

$$y''' + 4y' = \cos(2t) - 1$$

Solution:

Find the roots of the characteristic equation.

$$r^3 + 4r = 0$$

$$r(r^2 + 4) = 0$$

$$r = 0, \ \pm 2i$$

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With one distinct root and a pair of complex conjugate roots, the complementary solution is

$$y_c(t) = c_1 + c_2\cos(2t) + c_3\sin(2t)$$

Our guess for the particular solution would be

$$y_p(t) = A\cos(2t) + B\sin(2t) + C$$

but all three of the terms in this guess overlap with a term in the complementary solution. Multiplying the particular solution by t eliminates the overlap in all three terms, so the particular solution and its derivatives will be

$$y_p(t) = t(A\cos(2t) + B\sin(2t) + C)$$

$$y_p'(t) = (A + 2Bt)\cos(2t) + (B - 2At)\sin(2t) + C$$

$$y_p''(t) = (-4A - 4Bt)\sin(2t) + (4B - 4At)\cos(2t)$$

$$y_p'''(t) = (-12A - 8Bt)\cos(2t) + (-12B + 8At)\sin(2t)$$

Plugging these into the differential equation gives

$$\begin{aligned}(-12A - 8Bt)\cos(2t) + (-12B + 8At)\sin(2t)\\ +4(A + 2Bt)\cos(2t) + 4(B - 2At)\sin(2t) + 4C\\ = \cos(2t) - 1\end{aligned}$$

$$-8A\cos(2t) - 8B\sin(2t) + 4C = \cos(2t) - 1$$

Equating coefficients gives us the system of equations

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$$-8A = 1$$

$$-8B = 0$$

$$4C = -1$$

From this system, we find $A = -1/8$, $B = 0$, and $C = -1/4$, so the particular solution will be

$$y_p(t) = -\frac{1}{8}t\cos(2t) - \frac{t}{4}$$

The general solution will be the sum of the complementary and particular solutions.

$$y(t) = y_c(t) + y_p(t)$$

$$y(t) = c_1 + c_2\cos(2t) + c_3\sin(2t) - \frac{1}{8}t\cos(2t) - \frac{t}{4}$$

6. Find the general solution to the differential equation.

$$y'''' + 2y'' + y = t^2 + 8\sin t$$

Solution:

Find the roots of the characteristic equation.

$$r^4 + 2r^2 + 1 = 0$$

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$$(r^2+1)^2=0$$

$$r^2+1=0$$

$$r^2=-1$$

$$r=\pm i$$

With one repeated pair of complex conjugate roots, the complementary solution is

$$y_c(t)=c_1\cos t+c_2\sin t+c_3t\cos t+c_4t\sin t$$

Our guess for the particular solution would be

$$y_p(t)=At^2+Bt+C+D\sin t+E\cos t$$

but $D\sin t$ and $E\cos t$ overlap with terms in the complementary solution, so we'll multiply them by t^2, and our guess for the particular solution and its derivatives will be

$$y_p(t)=At^2+Bt+C+Dt^2\sin t+Et^2\cos t$$

$$y_p'(t)=2At+B+2t(D\sin t+E\cos t)+t^2(D\cos t-E\sin t)$$

$$y_p''(t)=2A+2(D\sin t+E\cos t)+4t(D\cos t-E\sin t)+t^2(-D\sin t-E\cos t)$$

$$y_p'''(t)=-6E\sin t+6D\cos t+6t(-D\sin t-E\cos t)+t^2(E\sin t-D\cos t)$$

$$y_p''''(t)=-12E\cos t-12D\sin t+8t(-D\cos t+E\sin t)+t^2(E\cos t+D\sin t)$$

Plugging these into the differential equation gives

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$$-12E\cos t - 12D\sin t + 8t(-D\cos t + E\sin t) + t^2(E\cos t + D\sin t)$$

$$+2[2A + 2(D\sin t + E\cos t) + 4t(D\cos t - E\sin t) + t^2(-D\sin t - E\cos t)]$$

$$+At^2 + Bt + C + Dt^2\sin t + Et^2\cos t = t^2 + 8\sin t$$

$$(-12E - 8Dt + Et^2 + 4E + 8Dt - 2Et^2 + Et^2)\cos t$$

$$+(-12D + 8Et + Dt^2 + 4D - 8Et - 2Dt^2 + Dt^2)\sin t$$

$$+At^2 + Bt + 4A + C = t^2 + 8\sin t$$

$$-8E\cos t - 8D\sin t + At^2 + Bt + 4A + C = t^2 + 8\sin t$$

Equating coefficients gives us the system of equations

$$A = 1$$

$$B = 0$$

$$4A + C = 0$$

$$-8D = 8$$

$$-8E = 0$$

From this system, we find $A = 1$, $B = 0$, $C = -4$, $D = -1$, and $E = 0$, so the particular solution will be

$$y_p(t) = t^2 - 4 - t^2\sin t$$

The general solution will be the sum of the complementary and particular solutions.

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$$y(t) = y_c(t) + y_p(t)$$

$$y(t) = c_1 \cos t + c_2 \sin t + c_3 t \cos t + c_4 t \sin t + t^2 - 4 - t^2 \sin t$$

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VARIATION OF PARAMETERS FOR HIGHER ORDER EQUATIONS

■ 1. Use variation of parameters to find the general solution to the differential equation.

$$y''' - y'' = te^t + 2t + 1 + 3\sin t$$

Solution:

Find the roots of the characteristic equation.

$$y''' - y'' = 0$$

$$r^3 - r^2 = 0$$

$$r^2(r - 1) = 0$$

$$r = 0,\ 0,\ 1$$

With a pair of repeated roots and one distinct root, the complementary solution is

$$y_c(t) = c_1e^{r_1t} + c_2te^{r_1t} + c_3e^{r_2t}$$

$$y_c(t) = c_1e^{0t} + c_2te^{0t} + c_3e^{1t}$$

$$y_c(t) = c_1 + c_2t + c_3e^t$$

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From the complementary solution, we can say that the fundamental set of solutions is $\{1, t, e^t\}$. With three solutions in the fundamental set, W, W_1, W_2, and W_3 are

$$W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 1 & t & e^t \\ 0 & 1 & e^t \\ 0 & 0 & e^t \end{vmatrix} = e^t$$

$$W_1 = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ 1 & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 0 & t & e^t \\ 0 & 1 & e^t \\ 1 & 0 & e^t \end{vmatrix} = 0 \begin{vmatrix} 1 & e^t \\ 0 & e^t \end{vmatrix} - t \begin{vmatrix} 0 & e^t \\ 1 & e^t \end{vmatrix} + e^t \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$$

$$= te^t - e^t = e^t(t-1)$$

$$W_2 = \begin{vmatrix} y_1 & 0 & y_3 \\ y_1' & 0 & y_3' \\ y_1'' & 1 & y_3'' \end{vmatrix} = \begin{vmatrix} 1 & 0 & e^t \\ 0 & 0 & e^t \\ 0 & 1 & e^t \end{vmatrix} = 1 \begin{vmatrix} 0 & e^t \\ 1 & e^t \end{vmatrix} - 0 \begin{vmatrix} 0 & e^t \\ 0 & e^t \end{vmatrix} + e^t \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} = -e^t$$

$$W_3 = \begin{vmatrix} y_1 & y_2 & 0 \\ y_1' & y_2' & 0 \\ y_1'' & y_2'' & 1 \end{vmatrix} = \begin{vmatrix} 1 & t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1$$

Now we'll use these values and $g(t)$ from the nonhomogeneous equation to find u_1', u_2', and u_3'. We'll get

$$u_1' = \frac{g(t)W_1}{W}$$

$$u_1' = \frac{(te^t + 2t + 1 + 3\sin t)(e^t(t-1))}{e^t}$$

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$$u_1' = (te^t + 2t + 1 + 3\sin t)(t - 1)$$

$$u_1' = t^2e^t + 2t^2 + t + 3t\sin t - te^t - 2t - 1 - 3\sin t$$

$$u_1' = t^2e^t + 2t^2 - t + 3t\sin t - te^t - 1 - 3\sin t$$

and

$$u_2' = \frac{g(t)W_2}{W}$$

$$u_2' = \frac{(te^t + 2t + 1 + 3\sin t)(-e^t)}{e^t}$$

$$u_2' = -te^t - 2t - 1 - 3\sin t$$

and

$$u_3' = \frac{g(t)W_3}{W}$$

$$u_3' = \frac{(te^t + 2t + 1 + 3\sin t)(1)}{e^t}$$

$$u_3' = t + 2te^{-t} + e^{-t} + 3e^{-t}\sin t$$

Use integration by parts to find u_1 from u_1'.

$$u_1 = \int t^2e^t + 2t^2 - t + 3t\sin t - te^t - 1 - 3\sin t\ dt$$

$$u_1 = t^2e^t - \int 2te^t\ dt + \frac{2}{3}t^3 - \frac{1}{2}t^2 - 3t\cos t + 3\int \cos t\ dt - \int te^t\ dt - t + 3\cos t$$

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$$u_1 = t^2e^t - 3te^t + 3\int e^t\ dt + \frac{2}{3}t^3 - \frac{1}{2}t^2 - 3t\cos t + 3\sin t - t + 3\cos t$$

$$u_1 = t^2e^t - 3te^t + 3e^t + \frac{2}{3}t^3 - \frac{1}{2}t^2 - t - 3t\cos t + 3\sin t + 3\cos t$$

Integrate u_2' to find u_2.

$$u_2 = \int -te^t - 2t - 1 - 3\sin t\ dt$$

$$u_2 = -te^t + \int e^t\ dt - t^2 - t + 3\cos t$$

$$u_2 = -te^t + e^t - t^2 - t + 3\cos t$$

Integrate u_3' to find u_3.

$$u_3 = \int t + 2te^{-t} + e^{-t} + 3e^{-t}\sin t\ dt$$

$$u_3 = \frac{1}{2}t^2 - e^{-t} + \int 2te^{-t}\ dt + \int 3e^{-t}\sin t\ dt$$

$$u_3 = \frac{1}{2}t^2 - e^{-t} + \left(-2te^{-t} - \int -2e^{-t}\ dt\right) + \int 3e^{-t}\sin t\ dt$$

$$u_3 = \frac{1}{2}t^2 - e^{-t} - 2te^{-t} + \int 2e^{-t}\ dt + \int 3e^{-t}\sin t\ dt$$

$$u_3 = \frac{1}{2}t^2 - e^{-t} - 2te^{-t} - 2e^{-t} + \int 3e^{-t}\sin t\ dt$$

$$u_3 = \frac{1}{2}t^2 - 3e^{-t} - 2te^{-t} + \int 3e^{-t}\sin t\ dt$$

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$$u_3 = \frac{1}{2}t^2 - 3e^{-t} - 2te^{-t} - \frac{3}{2}e^{-t}\cos t - \frac{3}{2}e^{-t}\sin t$$

$$u_3 = \frac{t^2}{2} - \frac{3}{e^t} - \frac{2t}{e^t} - \frac{3\cos t}{2e^t} - \frac{3\sin t}{2e^t}$$

With u_1, u_2, and u_3, we can say that the particular solution is

$$y_p(t) = u_1y_1 + u_2y_2 + u_3y_3$$

$$y_p(t) = \left(t^2e^t - 3te^t + 3e^t + \frac{2}{3}t^3 - \frac{1}{2}t^2 - t - 3t\cos t + 3\sin t + 3\cos t\right)(1)$$

$$+(-te^t + e^t - t^2 - t + 3\cos t)(t) + \left(\frac{t^2}{2} - \frac{3}{e^t} - \frac{2t}{e^t} - \frac{3\cos t}{2e^t} - \frac{3\sin t}{2e^t}\right)(e^t)$$

$$y_p(t) = t^2e^t - 3te^t + 3e^t + \frac{2}{3}t^3 - \frac{1}{2}t^2 - t - 3t\cos t + 3\sin t + 3\cos t$$

$$-t^2e^t + te^t - t^3 - t^2 + 3t\cos t + \frac{1}{2}t^2e^t - 3 - 2t - \frac{3\cos t}{2} - \frac{3\sin t}{2}$$

$$y_p(t) = -2te^t + 3e^t + \frac{1}{2}t^2e^t - \frac{1}{3}t^3 - \frac{3}{2}t^2 - 3t - 3 + \frac{3}{2}\sin t + \frac{3}{2}\cos t$$

Adding this particular solution to the complementary solution gives us the general solution $y(t)$.

$$y(t) = c_1 + c_2t + c_3e^t - 2te^t + 3e^t + \frac{t^2}{2}e^{-t} - \frac{1}{3}t^3 - \frac{3}{2}t^2 - 3t - 3 + \frac{3}{2}\sin t + \frac{3}{2}\cos t$$

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■ 2. Use variation of parameters to find the general solution to the differential equation.

$$y'' - 5y' + 6y = te^{2t} - e^{3t}$$

Solution:

Find the roots of the characteristic equation.

$$y'' - 5y' + 6y = 0$$

$$r^2 - 5r + 6 = 0$$

$$(r - 2)(r - 3) = 0$$

$$r = 2,\ 3$$

With two distinct roots, the complementary solution is

$$y_c(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$

$$y_c(t) = c_1 e^{2t} + c_2 e^{3t}$$

From the complementary solution, we can say that the fundamental set of solutions is $\{e^{2t}, e^{3t}\}$. Then W, W_1, and W_2 are

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{2t} & e^{3t} \\ 2e^{2t} & 3e^{3t} \end{vmatrix} = (e^{2t})(3e^{3t}) - (e^{3t})(2e^{2t}) = e^{5t}$$

$$W_1 = \begin{vmatrix} 0 & y_2 \\ 1 & y_2' \end{vmatrix} = \begin{vmatrix} 0 & e^{3t} \\ 1 & 3e^{3t} \end{vmatrix} = (0)(3e^{3t}) - (1)(e^{3t}) = -e^{3t}$$

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$$W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & 1 \end{vmatrix} = \begin{vmatrix} e^{2t} & 0 \\ 2e^{2t} & 1 \end{vmatrix} = (e^{2t})(1) - (0)(2e^{2t}) = e^{2t}$$

Now we'll use these values and $g(t)$ from the nonhomogeneous equation to find u_1' and u_2'.

$$u_1' = \frac{g(t)W_1}{W} = \frac{(te^{2t} - e^{3t})(-e^{3t})}{e^{5t}} = -t + e^t$$

$$u_2' = \frac{g(t)W_2}{W} = \frac{(te^{2t} - e^{3t})(e^{2t})}{e^{5t}} = te^{-t} - 1$$

Integrate u_1' to find u_1.

$$u_1 = \int (-t + e^t)\, dt$$

$$u_1 = -\frac{1}{2}t^2 + e^t$$

Integrate u_2' to find u_2.

$$u_2 = \int (te^{-t} - 1)\, dt$$

$$u_2 = -te^{-t} + \int e^{-t}\, dt - t$$

$$u_2 = -te^{-t} - e^{-t} - t$$

With u_1 and u_2, we can say that the particular solution is

$$y_p(t) = u_1 y_1 + u_2 y_2$$

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$$y_p(t) = \left(-\frac{1}{2}t^2 + e^t\right)(e^{2t}) + (-te^{-t} - e^{-t} - t)(e^{3t})$$

$$y_p(t) = -\frac{1}{2}t^2e^{2t} + e^{3t} - te^{2t} - e^{2t} - te^{3t}$$

Adding this particular solution to the complementary solution gives us the general solution $y(t)$.

$$y(t) = c_1e^{2t} + c_2e^{3t} - \frac{1}{2}t^2e^{2t} + e^{3t} - te^{2t} - e^{2t} - te^{3t}$$

■ 3. Use variation of parameters to find the general solution to the differential equation.

$$y''' - y'' - 6y' = t^2e^{3t} - t + 1$$

Solution:

Find the roots of the characteristic equation.

$$y''' - y'' - 6y' = 0$$

$$r^3 - r^2 - 6r = 0$$

$$r(r + 2)(r - 3) = 0$$

$$r = 0,\ -2,\ 3$$

With three distinct roots, the complementary solution is

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$$y_c(t) = c_1e^{r_1t} + c_2e^{r_1t} + c_3e^{r_2t}$$

$$y_c(t) = c_1e^{0t} + c_2e^{-2t} + c_3e^{3t}$$

$$y_c(t) = c_1 + c_2e^{-2t} + c_3e^{3t}$$

From the complementary solution, we can say that the fundamental set of solutions is $\{1, e^{-2t}, e^{3t}\}$. Then W, W_1, W_2, and W_3 are

$$W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 1 & e^{-2t} & e^{3t} \\ 0 & -2e^{-2t} & 3e^{3t} \\ 0 & 4e^{-2t} & 9e^{3t} \end{vmatrix} = -30e^t$$

$$W_1 = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ 1 & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 0 & e^{-2t} & e^{3t} \\ 0 & -2e^{-2t} & 3e^{3t} \\ 1 & 4e^{-2t} & 9e^{3t} \end{vmatrix} = 5e^t$$

$$W_2 = \begin{vmatrix} y_1 & 0 & y_3 \\ y_1' & 0 & y_3' \\ y_1'' & 1 & y_3'' \end{vmatrix} = \begin{vmatrix} 1 & 0 & e^{3t} \\ 0 & 0 & 3e^{3t} \\ 0 & 1 & 9e^{3t} \end{vmatrix} = -3e^{3t}$$

$$W_3 = \begin{vmatrix} y_1 & y_2 & 0 \\ y_1' & y_2' & 0 \\ y_1'' & y_2'' & 1 \end{vmatrix} = \begin{vmatrix} 1 & e^{-2t} & 0 \\ 0 & -2e^{-2t} & 0 \\ 0 & 4e^{-2t} & 1 \end{vmatrix} = -2e^{-2t}$$

Now we'll use these values and $g(t)$ from the nonhomogeneous equation to find u_1', u_2', and u_3'.

$$u_1' = \frac{g(t)W_1}{W}$$

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$$u_1' = \frac{(t^2e^{3t} - t + 1)(5e^t)}{-30e^t}$$

$$u_1' = -\frac{1}{6}t^2e^{3t} + \frac{1}{6}t - \frac{1}{6}$$

and

$$u_2' = \frac{g(t)W_2}{W}$$

$$u_2' = \frac{(t^2e^{3t} - t + 1)(-3e^{3t})}{-30e^t}$$

$$u_2' = \frac{1}{10}t^2e^{5t} - \frac{1}{10}te^{2t} + \frac{1}{10}e^{2t}$$

and

$$u_3' = \frac{g(t)W_3}{W}$$

$$u_3' = \frac{(t^2e^{3t} - t + 1)(-2e^{-2t})}{-30e^t}$$

$$u_3' = \frac{1}{15}t^2 - \frac{1}{15}te^{-3t} + \frac{1}{15}e^{-3t}$$

Use integration by parts to find u_1 from u_1'.

$$u_1 = \int -\frac{1}{6}t^2e^{3t} + \frac{1}{6}t - \frac{1}{6}\ dt$$

$$u_1 = -\frac{1}{18}t^2e^{3t} + \frac{1}{9}\int te^{3t}\ dt + \frac{1}{12}t^2 - \frac{1}{6}t$$

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$$u_1 = -\frac{1}{18}t^2e^{3t} + \frac{1}{27}te^{3t} - \frac{1}{27}\int e^{3t}\,dt + \frac{1}{12}t^2 - \frac{1}{6}t$$

$$u_1 = -\frac{1}{18}t^2e^{3t} + \frac{1}{27}te^{3t} - \frac{1}{81}e^{3t} + \frac{1}{12}t^2 - \frac{1}{6}t$$

Integrate u_2' to find u_2.

$$u_2 = \int \frac{1}{10}t^2e^{5t} - \frac{1}{10}te^{2t} + \frac{1}{10}e^{2t}\,dt$$

$$u_2 = \frac{1}{50}t^2e^{5t} - \frac{1}{25}\int te^{5t}\,dt - \frac{1}{20}te^{2t} + \frac{1}{20}\int e^{2t}\,dt + \frac{1}{20}e^{2t}$$

$$u_2 = \frac{1}{50}t^2e^{5t} - \frac{1}{125}te^{5t} + \frac{1}{125}\int e^{5t}\,dt - \frac{1}{20}te^{2t} + \frac{1}{40}e^{2t} + \frac{1}{20}e^{2t}$$

$$u_2 = \frac{1}{50}t^2e^{5t} - \frac{1}{125}te^{5t} + \frac{1}{625}e^{5t} - \frac{1}{20}te^{2t} + \frac{3}{40}e^{2t}$$

Integrate u_3' to find u_3.

$$u_3 = \int \frac{1}{15}t^2 - \frac{1}{15}te^{-3t} + \frac{1}{15}e^{-3t}\,dt$$

$$u_3 = \frac{1}{45}t^3 + \frac{1}{45}te^{-3t} - \frac{1}{45}\int e^{-3t}\,dt - \frac{1}{45}e^{-3t}$$

$$u_3 = \frac{1}{45}t^3 + \frac{1}{45}te^{-3t} + \frac{1}{135}e^{-3t} - \frac{1}{45}e^{-3t}$$

$$u_3 = \frac{1}{45}t^3 + \frac{1}{45}te^{-3t} - \frac{2}{135}e^{-3t}$$

With u_1, u_2, and u_3, we can say that the particular solution is

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$$y_p(t) = u_1y_1 + u_2y_2 + u_3y_3$$

$$y_p(t) = \left(-\frac{1}{18}t^2e^{3t} + \frac{1}{27}te^{3t} - \frac{1}{81}e^{3t} + \frac{1}{12}t^2 - \frac{1}{6}t\right)(1)$$

$$+\left(\frac{1}{50}t^2e^{5t} - \frac{1}{125}te^{5t} + \frac{1}{625}e^{5t} - \frac{1}{20}te^{2t} + \frac{3}{40}e^{2t}\right)e^{-2t}$$

$$+\left(\frac{1}{45}t^3 + \frac{1}{45}te^{-3t} - \frac{2}{135}e^{-3t}\right)e^{3t}$$

$$y_p(t) = -\frac{1}{18}t^2e^{3t} + \frac{1}{27}te^{3t} - \frac{1}{81}e^{3t} + \frac{1}{12}t^2 - \frac{1}{6}t$$

$$+\frac{1}{50}t^2e^{3t} - \frac{1}{125}te^{3t} + \frac{1}{625}e^{3t} - \frac{1}{20}t + \frac{3}{40} + \frac{1}{45}t^3e^{3t} + \frac{1}{45}t - \frac{2}{135}$$

$$y_p(t) = \frac{1}{45}t^3e^{3t} - \frac{8}{225}t^2e^{3t} + \frac{98}{3{,}375}te^{3t} - \frac{544}{50{,}625}e^{3t} + \frac{1}{12}t^2 - \frac{7}{36}t + \frac{13}{216}$$

Adding this particular solution to the complementary solution gives us the general solution $y(t)$.

$$y(t) = c_1 + c_2e^{-2t} + c_3e^{3t}$$

$$+\frac{1}{45}t^3e^{3t} - \frac{8}{225}t^2e^{3t} + \frac{98}{3{,}375}te^{3t} - \frac{544}{50{,}625}e^{3t} + \frac{1}{12}t^2 - \frac{7}{36}t + \frac{13}{216}$$

■ 4. Use variation of parameters to find the general solution to the differential equation.

$$y''' - 6y'' + 5y' + 12y = 10te^{3t} - 4\cos(2t)$$

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Solution:

Find the roots of the characteristic equation.

$$y''' - 6y'' + 5y' + 12y = 0$$

$$r^3 - 6r^2 + 5r + 12 = 0$$

$$(r + 1)(r - 3)(r - 4) = 0$$

$$r = -1, 3, 4$$

With three distinct roots, the complementary solution is

$$y_c(t) = c_1 e^{r_1 t} + c_2 e^{r_1 t} + c_3 e^{r_2 t}$$

$$y_c(t) = c_1 e^{-t} + c_2 e^{3t} + c_3 e^{4t}$$

From the complementary solution, we can say that the fundamental set of solutions is $\{e^{-t}, e^{3t}, e^{4t}\}$. Then W, W_1, W_2, and W_3 are

$$W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} e^{-t} & e^{3t} & e^{4t} \\ -e^{-t} & 3e^{3t} & 4e^{4t} \\ e^{-t} & 9e^{3t} & 16e^{4t} \end{vmatrix}$$

$$= e^{-t} \begin{vmatrix} 3e^{3t} & 4e^{4t} \\ 9e^{3t} & 16e^{4t} \end{vmatrix} - e^{3t} \begin{vmatrix} -e^{-t} & 4e^{4t} \\ e^{-t} & 16e^{4t} \end{vmatrix} + e^{4t} \begin{vmatrix} -e^{-t} & 3e^{3t} \\ e^{-t} & 9e^{3t} \end{vmatrix}$$

$$= e^{-t}(48e^{7t} - 36e^{7t}) - e^{3t}(-16e^{3t} - 4e^{3t}) + e^{4t}(-9e^{2t} - 3e^{2t})$$

$$= e^{-t}(12e^{7t}) - e^{3t}(-20e^{3t}) + e^{4t}(-12e^{2t})$$

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$$= 12e^{6t} + 20e^{6t} - 12e^{6t} = 20e^{6t}$$

$$W_1 = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ 1 & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 0 & e^{3t} & e^{4t} \\ 0 & 3e^{3t} & 4e^{4t} \\ 1 & 9e^{3t} & 16e^{4t} \end{vmatrix}$$

$$= 0 \begin{vmatrix} 3e^{3t} & 4e^{4t} \\ 9e^{3t} & 16e^{4t} \end{vmatrix} - e^{3t} \begin{vmatrix} 0 & 4e^{4t} \\ 1 & 16e^{4t} \end{vmatrix} + e^{4t} \begin{vmatrix} 0 & 3e^{3t} \\ 1 & 9e^{3t} \end{vmatrix}$$

$$= 0 - e^{3t}(0 - 4e^{4t}) + e^{4t}(0 - 3e^{3t})$$

$$= 4e^{7t} - 3e^{7t} = e^{7t}$$

$$W_2 = \begin{vmatrix} y_1 & 0 & y_3 \\ y_1' & 0 & y_3' \\ y_1'' & 1 & y_3'' \end{vmatrix} = \begin{vmatrix} e^{-t} & 0 & e^{4t} \\ -e^{-t} & 0 & 4e^{4t} \\ e^{-t} & 1 & 16e^{4t} \end{vmatrix}$$

$$= e^{-t} \begin{vmatrix} 0 & 4e^{4t} \\ 1 & 16e^{4t} \end{vmatrix} - 0 \begin{vmatrix} -e^{-t} & 4e^{4t} \\ e^{-t} & 16e^{4t} \end{vmatrix} + e^{4t} \begin{vmatrix} -e^{-t} & 0 \\ e^{-t} & 1 \end{vmatrix}$$

$$= e^{-t}(0 - 4e^{4t}) - 0 + e^{4t}(-e^{-t} - 0)$$

$$= -4e^{3t} - e^{3t} = -5e^{3t}$$

$$W_3 = \begin{vmatrix} y_1 & y_2 & 0 \\ y_1' & y_2' & 0 \\ y_1'' & y_2'' & 1 \end{vmatrix} = \begin{vmatrix} e^{-t} & e^{3t} & 0 \\ -e^{-t} & 3e^{3t} & 0 \\ e^{-t} & 9e^{3t} & 1 \end{vmatrix}$$

$$= e^{-t} \begin{vmatrix} 3e^{3t} & 0 \\ 9e^{3t} & 1 \end{vmatrix} - e^{3t} \begin{vmatrix} -e^{-t} & 0 \\ e^{-t} & 1 \end{vmatrix} + 0 \begin{vmatrix} -e^{-t} & 3e^{3t} \\ e^{-t} & 9e^{3t} \end{vmatrix}$$

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$$= e^{-t}(3e^{3t} - 0) - e^{3t}(-e^{-t} - 0) + 0$$

$$= 3e^{2t} + e^{2t} = 4e^{2t}$$

Now we'll use these values and $g(t)$ from the nonhomogeneous equation to find u_1', u_2', and u_3'.

$$u_1' = \frac{g(t)W_1}{W}$$

$$u_1' = \frac{(10te^{3t} - 4\cos(2t))(e^{7t})}{20e^{6t}}$$

$$u_1' = \frac{1}{2}te^{4t} - \frac{1}{5}e^{t}\cos(2t)$$

and

$$u_2' = \frac{g(t)W_2}{W}$$

$$u_2' = \frac{(10te^{3t} - 4\cos(2t))(-5e^{3t})}{20e^{6t}}$$

$$u_2' = -\frac{5}{2}t + e^{-3t}\cos(2t)$$

and

$$u_3' = \frac{g(t)W_3}{W}$$

$$u_3' = \frac{(10te^{3t} - 4\cos(2t))(4e^{2t})}{20e^{6t}}$$

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$$u_3' = 2te^{-t} - \frac{4}{5}e^{-4t}\cos(2t)$$

Use integration by parts to find u_1 from u_1'.

$$u_1 = \int \frac{1}{2}te^{4t} - \frac{1}{5}e^{t}\cos(2t)\ dt$$

$$u_1 = \frac{1}{8}te^{4t} - \frac{1}{8}\int e^{4t}\ dt - \frac{1}{5}\int e^{t}\cos(2t)\ dt$$

$$u_1 = \frac{1}{8}te^{4t} - \frac{1}{32}e^{4t} - \frac{1}{5}\int e^{t}\cos(2t)\ dt$$

$$\int e^{t}\cos(2t)\ dt = e^{t}\cos(2t) + 2\int e^{t}\sin(2t)\ dt$$

$$\int e^{t}\cos(2t)\ dt = e^{t}\cos(2t) + 2e^{t}\sin(2t) - 4\int e^{t}\cos(2t)\ dt$$

$$5\int e^{t}\cos(2t)\ dt = e^{t}\cos(2t) + 2e^{t}\sin(2t)$$

$$\int e^{t}\cos(2t)\ dt = \frac{1}{5}e^{t}\cos(2t) + \frac{2}{5}e^{t}\sin(2t)$$

$$u_1 = \frac{1}{8}te^{4t} - \frac{1}{32}e^{4t} - \frac{1}{25}e^{t}\cos(2t) - \frac{2}{25}e^{t}\sin(2t)$$

Use integration by parts to find u_2 from u_2'.

$$u_2 = \int -\frac{5}{2}t + e^{-3t}\cos(2t)\ dt$$

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$$u_2 = -\frac{5}{4}t^2 + \int e^{-3t}\cos(2t)\ dt$$

$$\int e^{-3t}\cos(2t)\ dt = -\frac{1}{3}e^{-3t}\cos(2t) - \frac{2}{3}\int e^{-3t}\sin(2t)\ dt$$

$$\int e^{-3t}\cos(2t)\ dt = -\frac{1}{3}e^{-3t}\cos(2t) + \frac{2}{9}e^{-3t}\sin(2t) - \frac{4}{9}\int e^{-3t}\cos(2t)\ dt$$

$$\frac{13}{9}\int e^{-3t}\cos(2t)\ dt = -\frac{1}{3}e^{-3t}\cos(2t) + \frac{2}{9}e^{-3t}\sin(2t)$$

$$\int e^{-3t}\cos(2t)\ dt = -\frac{3}{13}e^{-3t}\cos(2t) + \frac{2}{13}e^{-3t}\sin(2t)$$

$$u_2 = -\frac{5}{4}t^2 - \frac{3}{13}e^{-3t}\cos(2t) + \frac{2}{13}e^{-3t}\sin(2t)$$

Use integration by parts to find u_3 from u_3'.

$$u_3 = \int 2te^{-t} - \frac{4}{5}e^{-4t}\cos(2t)\ dt$$

$$u_3 = -2te^{-t} + 2\int e^{-t}\ dt - \frac{4}{5}\int e^{-4t}\cos(2t)\ dt$$

$$u_3 = -2te^{-t} - 2e^{-t} - \frac{4}{5}\int e^{-4t}\cos(2t)\ dt$$

$$\int e^{-4t}\cos(2t)\ dt = -\frac{1}{4}e^{-4t}\cos(2t) - \frac{1}{2}\int e^{-4t}\sin(2t)\ dt$$

$$\int e^{-4t}\cos(2t)\ dt = -\frac{1}{4}e^{-4t}\cos(2t) + \frac{1}{8}e^{-4t}\sin(2t) - \frac{1}{4}\int e^{-4t}\cos(2t)\ dt$$

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$$\frac{5}{4}\int e^{-4t}\cos(2t)\,dt = -\frac{1}{4}e^{-4t}\cos(2t) + \frac{1}{8}e^{-4t}\sin(2t)$$

$$\int e^{-4t}\cos(2t)\,dt = -\frac{1}{5}e^{-4t}\cos(2t) + \frac{1}{10}e^{-4t}\sin(2t)$$

$$u_3 = -2te^{-t} - 2e^{-t} + \frac{4}{25}e^{-4t}\cos(2t) - \frac{2}{25}e^{-4t}\sin(2t)$$

With u_1, u_2, and u_3, we can say that the particular solution is

$$y_p(t) = u_1y_1 + u_2y_2 + u_3y_3$$

$$y_p(t) = \left(\frac{1}{8}te^{4t} - \frac{1}{32}e^{4t} - \frac{1}{25}e^{t}\cos(2t) - \frac{2}{25}e^{t}\sin(2t)\right)(e^{-t})$$

$$+\left(-\frac{5}{4}t^2 - \frac{3}{13}e^{-3t}\cos(2t) + \frac{2}{13}e^{-3t}\sin(2t)\right)e^{3t}$$

$$+\left(-2te^{-t} - 2e^{-t} + \frac{4}{25}e^{-4t}\cos(2t) - \frac{2}{25}e^{-4t}\sin(2t)\right)e^{4t}$$

$$y_p(t) = \frac{1}{8}te^{3t} - \frac{1}{32}e^{3t} - \frac{1}{25}\cos(2t) - \frac{2}{25}\sin(2t)$$

$$-\frac{5}{4}t^2e^{3t} - \frac{3}{13}\cos(2t) + \frac{2}{13}\sin(2t) - 2te^{3t} - 2e^{3t} + \frac{4}{25}\cos(2t) - \frac{2}{25}\sin(2t)$$

$$y_p(t) = -\frac{5}{4}t^2e^{3t} - \frac{15}{8}te^{3t} - \frac{65}{32}e^{3t} - \frac{36}{325}\cos(2t) - \frac{2}{325}\sin(2t)$$

Adding this particular solution to the complementary solution gives us the general solution $y(t)$.

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$$y(t) = c_1e^{-t} + c_2e3t + c_3e^{4t} - \frac{5}{4}t^2e^{3t} - \frac{15}{8}te^{3t} - \frac{65}{32}e^{3t}$$

$$-\frac{36}{325}\cos(2t) - \frac{2}{325}\sin(2t)$$

■ 5. Use variation of parameters to find the general solution to the differential equation.

$$y''' - y'' + 9y' - 9y = 60\cos(3t)$$

Solution:

Find the roots of the characteristic equation.

$$y''' - y'' + 9y' - 9y = 0$$

$$r^3 - r^2 + 9r - 9 = 0$$

$$(r - 1)(r^2 + 9) = 0$$

$$r = 1, \ \pm 3i$$

With a pair of complex conjugate roots and one distinct root, the complementary solution is

$$y_c(t) = c_1e^t + c_2\cos(3t) + c_3\sin(3t)$$

From the complementary solution, we can say that the fundamental set of solutions is $\{e^t, \cos(3t), \sin(3t)\}$. Then W, W_1, W_2, and W_3 are

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$$W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} e^t & \cos(3t) & \sin(3t) \\ e^t & -3\sin(3t) & 3\cos(3t) \\ e^t & -9\cos(3t) & -9\sin(3t) \end{vmatrix}$$

$$= e^t \begin{vmatrix} -3\sin(3t) & 3\cos(3t) \\ -9\cos(3t) & -9\sin(3t) \end{vmatrix} - \cos(3t) \begin{vmatrix} e^t & 3\cos(3t) \\ e^t & -9\sin(3t) \end{vmatrix}$$

$$+\sin(3t) \begin{vmatrix} e^t & -3\sin(3t) \\ e^t & -9\cos(3t) \end{vmatrix}$$

$$= e^t(27\sin^2(3t) + 27\cos^2(3t)) - \cos(3t)(-9e^t\sin(3t) - 3e^t\cos(3t))$$

$$+\sin(3t)(-9e^t\cos(3t) + 3e^t\sin(3t))$$

$$= 27e^t + 9e^t\sin(3t)\cos(3t) + 3e^t\cos^2(3t) - 9e^t\sin(3t)\cos(3t) + 3e^t\sin^2(3t)$$

$$= 27e^t + 3e^t$$

$$= 30e^t$$

$$W_1 = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ 1 & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 0 & \cos(3t) & \sin(3t) \\ 0 & -3\sin(3t) & 3\cos(3t) \\ 1 & -9\cos(3t) & -9\sin(3t) \end{vmatrix}$$

$$= 0 \begin{vmatrix} -3\sin(3t) & 3\cos(3t) \\ -9\cos(3t) & -9\sin(3t) \end{vmatrix} - \cos(3t) \begin{vmatrix} 0 & 3\cos(3t) \\ 1 & -9\sin(3t) \end{vmatrix}$$

$$+\sin(3t) \begin{vmatrix} 0 & -3\sin(3t) \\ 1 & -9\cos(3t) \end{vmatrix}$$

$$= 0 - \cos(3t)(-3\cos(3t)) + \sin(3t)(3\sin 3(3t)t)$$

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$$= 3\cos^2(3t) + 3\sin^2(3t)$$

$$= 3$$

$$W_2 = \begin{vmatrix} y_1 & 0 & y_3 \\ y_1' & 0 & y_3' \\ y_1'' & 1 & y_3'' \end{vmatrix} = \begin{vmatrix} e^t & 0 & \sin(3t) \\ e^t & 0 & 3\cos(3t) \\ e^t & 1 & -9\sin(3t) \end{vmatrix}$$

$$= e^t \begin{vmatrix} 0 & 3\cos(3t) \\ 1 & -9\sin(3t) \end{vmatrix} - 0 \begin{vmatrix} e^t & 3\cos(3t) \\ e^t & -9\sin(3t) \end{vmatrix} + \sin(3t) \begin{vmatrix} e^t & 0 \\ e^t & 1 \end{vmatrix}$$

$$= e^t(-3\cos(3t)) - 0 + \sin(3t)(e^t)$$

$$= -3e^t\cos(3t) + e^t\sin(3t)$$

$$W_3 = \begin{vmatrix} y_1 & y_2 & 0 \\ y_1' & y_2' & 0 \\ y_1'' & y_2'' & 1 \end{vmatrix} = \begin{vmatrix} e^t & \cos(3t) & 0 \\ e^t & -3\sin(3t) & 0 \\ e^t & -9\cos(3t) & 1 \end{vmatrix}$$

$$= e^t \begin{vmatrix} -3\sin(3t) & 0 \\ -9\cos(3t) & 1 \end{vmatrix} - \cos(3t) \begin{vmatrix} e^t & 0 \\ e^t & 1 \end{vmatrix} + 0 \begin{vmatrix} e^t & -3\sin(3t) \\ e^t & -9\cos(3t) \end{vmatrix}$$

$$= e^t(-3\sin(3t)) - \cos(3t)(e^t) + 0$$

$$= -3e^t\sin(3t) - e^t\cos(3t)$$

Now we'll use these values and $g(t)$ from the nonhomogeneous equation to find u_1', u_2', and u_3'.

$$u_1' = \frac{g(t)W_1}{W}$$

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$$u_1' = \frac{3(60\cos(3t))}{30e^t}$$

$$u_1' = 6e^{-t}\cos(3t)$$

and

$$u_2' = \frac{g(t)W_2}{W}$$

$$u_2' = \frac{(60\cos(3t))(-3e^t\cos(3t) + e^t\sin(3t))}{30e^t}$$

$$u_2' = -6\cos^2(3t) + 2\sin(3t)\cos(3t)$$

$$u_2' = -3 - 3\cos(6t) + \sin(6t)$$

and

$$u_3' = \frac{g(t)W_3}{W}$$

$$u_3' = \frac{(60\cos(3t))(-3e^t\sin(3t) - e^t\cos(3t))}{30e^t}$$

$$u_3' = -6\sin(3t)\cos(3t) - 2\cos^2(3t)$$

$$u_3' = -3\sin(6t) - 1 - \cos(6t)$$

Use integration by parts to find u_1 from u_1'.

$$u_1 = \int 6e^{-t}\cos(3t)\ dt = 6\int e^{-t}\cos(3t)\ dt$$

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$$\int e^{-t}\cos(3t)\ dt = -e^{-t}\cos(3t) - 3\int e^{-t}\sin(3t)\ dt$$

$$\int e^{-t}\cos(3t)\ dt = -e^{-t}\cos(3t) + 3e^{-t}\sin(3t) - 9\int e^{-t}\cos(3t)\ dt$$

$$10\int e^{-t}\cos(3t)\ dt = -e^{-t}\cos(3t) + 3e^{-t}\sin(3t)$$

$$\int e^{-t}\cos(3t)\ dt = -\frac{1}{10}e^{-t}\cos(3t) + \frac{3}{10}e^{-t}\sin(3t)$$

$$u_1 = 6\left(-\frac{1}{10}e^{-t}\cos(3t) + \frac{3}{10}e^{-t}\sin(3t)\right)$$

$$u_1 = -\frac{3}{5}e^{-t}\cos(3t) + \frac{9}{5}e^{-t}\sin(3t)$$

Integrate u_2' to find u_2.

$$u_2 = \int -3 - 3\cos(6t) + \sin(6t)\ dt$$

$$u_2 = -3t - \frac{1}{2}\sin(6t) - \frac{1}{6}\cos(6t)$$

Integrate u_3' to find u_3.

$$u_3 = \int -3\sin(6t) - 1 - \cos(6t)\ dt$$

$$u_3 = \frac{1}{2}\cos(6t) - t - \frac{1}{6}\sin(6t)$$

With u_1, u_2, and u_3, we can say that the particular solution is

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$$y_p(t) = u_1y_1 + u_2y_2 + u_3y_3$$

$$y_p(t) = \left(-\frac{3}{5}e^{-t}\cos(3t) + \frac{9}{5}e^{-t}\sin(3t)\right)e^t + \left(-3t - \frac{1}{2}\sin(6t) - \frac{1}{6}\cos(6t)\right)\cos(3t)$$

$$+\left(\frac{1}{2}\cos(6t) - t - \frac{1}{6}\sin(6t)\right)\sin(3t)$$

$$y_p(t) = -\frac{3}{5}\cos(3t) + \frac{9}{5}\sin(3t) + \left(-3t - \frac{1}{2}\sin(6t) - \frac{1}{6}\cos(6t)\right)\cos(3t)$$

$$+\left(\frac{1}{2}\cos(6t) - t - \frac{1}{6}\sin(6t)\right)\sin(3t)$$

Adding this particular solution to the complementary solution gives us the general solution $y(t)$.

$$y(t) = c_1e^t + c_2\cos(3t) + c_3\sin(3t) - \frac{3}{5}\cos(3t) + \frac{9}{5}\sin(3t)$$

$$+\left(-3t - \frac{1}{2}\sin(6t) - \frac{1}{6}\cos(6t)\right)\cos(3t)$$

$$+\left(\frac{1}{2}\cos(6t) - t - \frac{1}{6}\sin(6t)\right)\sin(3t)$$

■ 6. Use variation of parameters to find the general solution to the differential equation, given $y(0) = 0$, $y'(0) = -1$, $y''(0) = 10$.

$$y''' + y'' - y' - y = 8e^{-t}$$

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Solution:

Find the roots of the characteristic equation.

$$y''' + y'' - y' - y = 0$$

$$r^3 + r^2 - r - 1 = 0$$

$$(r + 1)(r^2 - 1) = 0$$

$$r = -1, \ -1, \ 1$$

With a pair of equal roots and one distinct root, the complementary solution is

$$y_c(t) = c_1 e^{r_1 t} + c_2 t e^{r_1 t} + c_3 e^{r_2 t}$$

$$y_c(t) = c_1 e^{-t} + c_2 t e^{-t} + c_3 e^{t}$$

From the complementary solution, we can say that the fundamental set of solutions is $\{e^{-t}, te^{-t}, e^{t}\}$. Then W, W_1, W_2, and W_3 are

$$W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} e^{-t} & te^{-t} & e^{t} \\ -e^{-t} & e^{-t}(1-t) & e^{t} \\ e^{-t} & e^{-t}(t-2) & e^{t} \end{vmatrix} = 4e^{-t}$$

$$W_1 = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ 1 & y_2'' & y_3'' \end{vmatrix} = \begin{vmatrix} 0 & te^{-t} & e^{t} \\ 0 & e^{-t}(1-t) & e^{t} \\ 1 & e^{-t}(t-2) & e^{t} \end{vmatrix} = 2t - 1$$

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$$W_2 = \begin{vmatrix} y_1 & 0 & y_3 \\ y_1' & 0 & y_3' \\ y_1'' & 1 & y_3'' \end{vmatrix} = \begin{vmatrix} e^{-t} & 0 & e^{t} \\ -e^{-t} & 0 & e^{t} \\ e^{-t} & 1 & e^{t} \end{vmatrix} = -2$$

$$W_3 = \begin{vmatrix} y_1 & y_2 & 0 \\ y_1' & y_2' & 0 \\ y_1'' & y_2'' & 1 \end{vmatrix} = \begin{vmatrix} e^{-t} & te^{-t} & 0 \\ -e^{-t} & e^{-t}(1-t) & 0 \\ e^{-t} & e^{-t}(t-2) & 1 \end{vmatrix} = e^{-2t}$$

Now we'll use these values and $g(t)$ from the nonhomogeneous equation to find u_1', u_2', and u_3'.

$$u_1' = \frac{g(t)W_1}{W} = \frac{8e^{-t}(2t-1)}{4e^{-t}} = 4t - 2$$

$$u_2' = \frac{g(t)W_2}{W} = \frac{8e^{-t}(-2)}{4e^{-t}} = -4$$

$$u_3' = \frac{g(t)W_3}{W} = \frac{8e^{-t}(e^{-2t})}{4e^{-t}} = 2e^{-2t}$$

Use integration by parts to find u_1 from u_1'.

$$u_1 = \int 4t - 2 \; dt$$

$$u_1 = -2t^2 - 2t$$

Integrate u_2' to find u_2.

$$u_2 = \int -4 \; dt$$

$$u_2 = -4t$$

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Integrate u_3' to find u_3.

$$u_3 = \int 2e^{-2t}\, dt$$

$$u_3 = -e^{-2t}$$

With u_1, u_2, and u_3, we can say that the particular solution is

$$y_p(t) = u_1y_1 + u_2y_2 + u_3y_3$$

$$y_p(t) = (2t^2 - 2t)e^{-t} - 4t(te^{-t}) - e^{-2t}e^t$$

$$y_p(t) = 2t^2e^{-t} - 2te^{-t} - 4t^2e^{-t} - e^{-t}$$

$$y_p(t) = -2t^2e^{-t} - 2te^{-t} - e^{-t}$$

Adding this particular solution to the complementary solution gives us the general solution $y(t)$.

$$y(t) = c_1e^{-t} + c_2te^{-t} + c_3e^t - 2t^2e^{-t} - 2te^{-t} - e^{-t}$$

The derivatives of the general solution are

$$y'(t) = -c_1e^{-t} + c_2e^{-t} - c_2te^{-t} + c_3e^t - 2te^{-t} + 2t^2e^{-t} - e^{-t}$$

$$y''(t) = c_1e^{-t} - 2c_2e^{-t} + c_2te^{-t} + c_3e^t - e^{-t} + 6te^{-t} - 2t^2e^{-t}$$

Substituting the initial conditions,

$$y(0) = c_1 + c_3 - 1 = 0$$

$$y'(0) = -c_1 + c_2 + c_3 - 1 = -1$$

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$$y''(0) = c_1 - 2c_2 + c_3 - 1 = 10$$

and then solving these as a system of equations gives $c_1 = 3$, $c_2 = 5$, and $c_3 = -2$. So the actual solution is

$$y(t) = 3e^{-t} + 5te^{-t} - 2e^{t} - 2t^2e^{-t} - 2te^{-t} - e^{-t}$$

$$y(t) = 2e^{-t} + 3te^{-t} - 2e^{t} - 2t^2e^{-t}$$

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LAPLACE TRANSFORMS FOR HIGHER ORDER EQUATIONS

■ 1. Use the Laplace transform to solve the differential equation, given $y(0) = 0$, $y'(0) = 0$, and $y''(0) = 0$.

$$y''' + 6y'' + 8y' = 8e^{-2t}$$

Solution:

Apply the Laplace transform to both sides of the differential equation.

$$\mathcal{L}(y''') + 6\mathcal{L}(y'') + 8\mathcal{L}(y') = 8\mathcal{L}(e^{-2t})$$

$$s^3Y(s) - s^2y(0) - sy'(0) - y''(0) + 6(s^2Y(s) - sy(0) - y'(0))$$

$$+8(sY(s) - y(0)) = \frac{8}{s+2}$$

Substitute the initial conditions $y(0) = 0$, $y'(0) = 0$, and $y''(0) = 0$,

$$Y(s)(s^3 + 6s^2 + 8s) = \frac{8}{s+2}$$

then solve for $Y(s)$.

$$Y(s) = \frac{8}{(s+2)(s^3 + 6s^2 + 8s)}$$

$$Y(s) = \frac{8}{s(s+2)^2(s+4)}$$

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Apply a partial fractions decomposition.

$$\frac{8}{s(s+2)^2(s+4)} = \frac{A}{s} + \frac{B}{(s+2)^2} + \frac{C}{s+2} + \frac{D}{s+4}$$

We can quickly find $A = 1/2$, $B = -2$, and $D = -1/2$, then we'll plug these values into the decomposition, along with $s = 1$, to solve for C.

$$\frac{8}{(1+2)^2(1+4)} = \frac{1}{2(1)} - \frac{2}{(1+2)^2} + \frac{C}{1+2} - \frac{1}{2(1+4)}$$

$$\frac{C}{3} = \frac{8}{45} - \frac{1}{2} + \frac{2}{9} + \frac{1}{10}$$

$$C = 0$$

Then the Laplace transform is

$$Y(s) = \frac{1}{2s} - \frac{2}{(s+2)^2} - \frac{1}{2(s+4)}$$

Applying the inverse Laplace transform to each term gives the general solution to the differential equation gives

$$y(t) = \frac{1}{2} - 2te^{-2t} - \frac{1}{2}e^{-4t}$$

■ 2. Use the Laplace transform to solve the differential equation, given $y(0) = 0$, $y'(0) = 0$, and $y''(0) = 1$.

$$y''' - 3y' + 2y = 12e^t + 24e^{2t}$$

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Solution:

Apply the Laplace transform to both sides of the differential equation.

$$\mathcal{L}(y''') - 3\mathcal{L}(y') + 2\mathcal{L}(y) = 12\mathcal{L}(e^t) + 24\mathcal{L}(e^{2t})$$

$$s^3Y(s) - s^2y(0) - sy'(0) - y''(0) - 3(sY(s) - y(0)) + 2Y(s) = \frac{12}{s-1} + \frac{24}{s-2}$$

Substitute the initial conditions $y(0) = 0$, $y'(0) = 0$, and $y''(0) = 1$,

$$Y(s)(s^3 - 3s + 2) = \frac{12}{s-1} + \frac{24}{s-2} + 1 = \frac{s^2 + 33s - 46}{(s-1)(s-2)}$$

then solve for $Y(s)$.

$$Y(s) = \frac{s^2 + 33s - 46}{(s-1)(s-2)(s-1)^2(s+2)} = \frac{s^2 + 33s - 46}{(s-1)^3(s-2)(s+2)}$$

Apply a partial fractions decomposition.

$$\frac{s^2 + 33s - 46}{(s-1)^3(s-2)(s+2)} = \frac{A}{(s-1)^3} + \frac{B}{(s-1)^2} + \frac{C}{s-1} + \frac{D}{s-2} + \frac{E}{s+2}$$

We can quickly find $A = 4$, $D = 6$, and $E = -1$, then we'll plug these values into the decomposition, along with $s = -1$,

$$\frac{(-1)^2 + 33(-1) - 46}{(-1-1)^3(-1-2)(-1+2)} = \frac{4}{(-1-1)^3} + \frac{B}{(-1-1)^2} + \frac{C}{-1-1} + \frac{6}{-1-2} - \frac{1}{-1+2}$$

$$B - 2C = 1$$

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and then separately $s = 0$,

$$\frac{0^2 + 33(0) - 46}{(0-1)^3(0-2)(0+2)} = \frac{4}{(0-1)^3} + \frac{B}{(0-1)^2} + \frac{C}{0-1} + \frac{6}{0-2} - \frac{1}{0+2}$$

$$B - C = -4$$

to get the system of equations

$$B - 2C = 1$$

$$B - C = -4$$

Solving this system gives $B = -9$ and $C = -5$. Then the Laplace transform is

$$Y(s) = \frac{4}{(s-1)^3} - \frac{9}{(s-1)^2} - \frac{5}{s-1} + \frac{6}{s-2} - \frac{1}{s+2}$$

Applying the inverse Laplace transform to each term gives the general solution to the differential equation gives

$$y(t) = \frac{4}{2!}t^2e^t - 9te^t - 5e^t + 6e^{2t} - e^{-2t}$$

$$y(t) = 2t^2e^t - 9te^t - 5e^t + 6e^{2t} - e^{-2t}$$

■ 3. Use the Laplace transform to solve the differential equation, given $y(0) = 0$, $y'(0) = 1$, and $y''(0) = 2$.

$$y''' - y'' - 5y' - 3y = 2e^{3t} + 10\sin t$$

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Solution:

Apply the Laplace transform to both sides of the differential equation.

$$\mathscr{L}(y''') - \mathscr{L}(y'') - 5\mathscr{L}(y') - 3\mathscr{L}(y) = 2\mathscr{L}(e^{3t}) + 10\mathscr{L}(\sin t)$$

$$s^3Y(s) - s^2y(0) - sy'(0) - y''(0) - s^2Y(s) + sy(0) + y'(0)$$

$$-5sY(s) + 5y(0) - 3Y(s) = \frac{2}{s-3} + \frac{10}{s^2+1}$$

Substitute the initial conditions $y(0) = 0$, $y'(0) = 1$, and $y''(0) = 2$,

$$Y(s)(s^3 - s^2 - 5s - 3) = \frac{2}{s-3} + \frac{10}{s^2+1} + s + 1$$

$$Y(s)(s^3 - s^2 - 5s - 3) = \frac{s^4 - 2s^3 + 8s - 31}{(s-3)(s^2+1)}$$

then solve for $Y(s)$.

$$Y(s) = \frac{s^4 - 2s^3 + 8s - 31}{(s-3)^2(s^2+1)(s+1)^2}$$

Apply a partial fractions decomposition.

$$\frac{s^4 - 2s^3 + 8s - 31}{(s^2+1)(s+1)^2(s-3)^2} = \frac{As+B}{s^2+1} + \frac{C}{(s+1)^2} + \frac{D}{s+1} + \frac{E}{(s-3)^2} + \frac{F}{s-3}$$

We can quickly find $C = -9/8$ and $E = 1/8$, then we'll plug these values into the decomposition, along with $s = -2$,

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$$\frac{(-2)^4-2(-2)^3+8(-2)-31}{((-2)^2+1)(-2+1)^2(-2-3)^2}$$

$$=\frac{A(-2)+B}{(-2)^2+1}-\frac{9}{8(-2+1)^2}+\frac{D}{-2+1}+\frac{1}{8(-2-3)^2}+\frac{F}{-2-3}$$

$$2A-B+5D+F=-5$$

and then separately $s=0$,

$$\frac{0^4-2(0)^3+8(0)-31}{(0^2+1)(0+1)^2(0-3)^2}=\frac{A(0)+B}{0^2+1}-\frac{9}{8(0+1)^2}+\frac{D}{0+1}+\frac{1}{8(0-3)^2}+\frac{F}{0-3}$$

$$3B+3D-F=-7$$

and then separately $s=1$,

$$\frac{1^4-2(1)^3+8(1)-31}{(1^2+1)(1+1)^2(1-3)^2}=\frac{A(1)+B}{1^2+1}-\frac{9}{8(1+1)^2}+\frac{D}{1+1}+\frac{1}{8(1-3)^2}+\frac{F}{1-3}$$

$$A+B+D-F=-1$$

and then separately $s=2$,

$$\frac{2^4-2(2)^3+8(2)-31}{(2^2+1)(2+1)^2(2-3)^2}=\frac{A(2)+B}{2^2+1}-\frac{9}{8(2+1)^2}+\frac{D}{2+1}+\frac{1}{8(2-3)^2}+\frac{F}{2-3}$$

$$6A+3B+5D-15F=-5$$

to get the system of equations

$$2A-B+5D+F=-5$$

$$3B+3D-F=-7$$

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$$A + B + D - F = -1$$

$$6A + 3B + 5D - 15F = -5$$

We can use a matrix to solve the system for the values of A, B, D, and F.

$$\left[\begin{array}{cccc|c} 2 & -1 & 5 & 1 & -5 \\ 0 & 3 & 3 & -1 & -7 \\ 1 & 1 & 1 & -1 & -1 \\ 6 & 3 & 5 & -15 & -5 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 1 & 1 & -1 & -1 \\ 2 & -1 & 5 & 1 & -5 \\ 0 & 3 & 3 & -1 & -7 \\ 6 & 3 & 5 & -15 & -5 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 1 & 1 & -1 & -1 \\ 0 & -3 & 3 & 3 & -3 \\ 0 & 3 & 3 & -1 & -7 \\ 6 & 3 & 5 & -15 & -5 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 1 & 1 & -1 & -1 \\ 0 & -3 & 3 & 3 & -3 \\ 0 & 3 & 3 & -1 & -7 \\ 0 & -3 & -1 & -9 & 1 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 1 & 1 & -1 & -1 \\ 0 & 1 & -1 & -1 & 1 \\ 0 & 3 & 3 & -1 & -7 \\ 0 & -3 & -1 & -9 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 2 & 0 & -2 \\ 0 & 1 & -1 & -1 & 1 \\ 0 & 3 & 3 & -1 & -7 \\ 0 & -3 & -1 & -9 & 1 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 2 & 0 & -2 \\ 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 6 & 2 & -10 \\ 0 & -3 & -1 & -9 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 2 & 0 & -2 \\ 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 6 & 2 & -10 \\ 0 & 0 & -4 & -12 & 4 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 2 & 0 & -2 \\ 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & \frac{1}{3} & -\frac{5}{3} \\ 0 & 0 & -4 & -12 & 4 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & -\frac{2}{3} & \frac{4}{3} \\ 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & \frac{1}{3} & -\frac{5}{3} \\ 0 & 0 & -4 & -12 & 4 \end{array}\right]$$

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$$\rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & -\frac{2}{3} & \frac{4}{3} \\ 0 & 1 & 0 & -\frac{2}{3} & -\frac{2}{3} \\ 0 & 0 & 1 & \frac{1}{3} & -\frac{5}{3} \\ 0 & 0 & -4 & -12 & 4 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & -\frac{2}{3} & \frac{4}{3} \\ 0 & 1 & 0 & -\frac{2}{3} & -\frac{2}{3} \\ 0 & 0 & 1 & \frac{1}{3} & -\frac{5}{3} \\ 0 & 0 & 0 & -\frac{32}{3} & -\frac{8}{3} \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & -\frac{2}{3} & \frac{4}{3} \\ 0 & 1 & 0 & -\frac{2}{3} & -\frac{2}{3} \\ 0 & 0 & 1 & \frac{1}{3} & -\frac{5}{3} \\ 0 & 0 & 0 & 1 & \frac{1}{4} \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & \frac{3}{2} \\ 0 & 1 & 0 & -\frac{2}{3} & -\frac{2}{3} \\ 0 & 0 & 1 & \frac{1}{3} & -\frac{5}{3} \\ 0 & 0 & 0 & 1 & \frac{1}{4} \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & \frac{3}{2} \\ 0 & 1 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{3} & -\frac{5}{3} \\ 0 & 0 & 0 & 1 & \frac{1}{4} \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & \frac{3}{2} \\ 0 & 1 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 1 & 0 & -\frac{7}{4} \\ 0 & 0 & 0 & 1 & \frac{1}{4} \end{array}\right]$$

Solving this system gives $A = 3/2$, $B = -1/2$, $D = -7/4$, and $F = 1/4$. Then the Laplace transform is

$$Y(s) = \frac{\frac{3}{2}s - \frac{1}{2}}{s^2 + 1} + \frac{-\frac{9}{8}}{(s+1)^2} + \frac{-\frac{7}{4}}{s+1} + \frac{\frac{1}{8}}{(s-3)^2} + \frac{\frac{1}{4}}{s-3}$$

$$Y(s) = \frac{1}{2}\left(\frac{3s-1}{s^2+1}\right) - \frac{9}{8}\left(\frac{1}{(s+1)^2}\right) - \frac{7}{4}\left(\frac{1}{s+1}\right)$$

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$$+\frac{1}{8}\left(\frac{1}{(s-3)^2}\right)+\frac{1}{4}\left(\frac{1}{s-3}\right)$$

$$Y(s)=\frac{3}{2}\left(\frac{s}{s^2+1^2}\right)-\frac{1}{2}\left(\frac{1}{s^2+1^2}\right)-\frac{9}{8}\left(\frac{1}{(s-(-1))^{1+1}}\right)-\frac{7}{4}\left(\frac{1}{s-(-1)}\right)$$

$$+\frac{1}{8}\left(\frac{1}{(s-3)^{1+1}}\right)+\frac{1}{4}\left(\frac{1}{s-3}\right)$$

Applying the inverse Laplace transform to each term gives the general solution to the differential equation gives

$$y(t)=\frac{3}{2}\cos t-\frac{1}{2}\sin t-\frac{9}{8}te^{-t}-\frac{7}{4}e^{-t}+\frac{1}{8}te^{3t}+\frac{1}{4}e^{3t}$$

4. Use the Laplace transform to solve the differential equation, given $y(0)=0$, $y'(0)=0$, and $y''(0)=0$.

$$y'''-2y''+y'-2y=5+25\sin t$$

Solution:

Apply the Laplace transform to both sides of the differential equation.

$$\mathscr{L}(y''')-2\mathscr{L}(y'')+\mathscr{L}(y')-2\mathscr{L}(y)=\mathscr{L}(5)+25\mathscr{L}(\sin t)$$

$$s^3Y(s)-s^2y(0)-sy'(0)-y''(0)-2s^2Y(s)+2sy(0)+2y'(0)$$

$$+sY(s)-y(0)-2Y(s)=\frac{5}{s}+\frac{25}{s^2+1}$$

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Substitute the initial conditions $y(0) = 0$, $y'(0) = 0$, and $y''(0) = 0$,

$$Y(s)(s^3 - 2s^2 + s - 2) = \frac{5}{s} + \frac{25}{s^2 + 1} = \frac{5s^2 + 25s + 5}{s(s^2 + 1)}$$

then solve for $Y(s)$.

$$Y(s) = \frac{5(s^2 + 5s + 1)}{s(s^2 + 1)^2(s - 2)}$$

Apply a partial fractions decomposition.

$$\frac{5(s^2 + 5s + 1)}{s(s - 2)(s^2 + 1)^2} = \frac{A}{s} + \frac{B}{s - 2} + \frac{Cs + D}{s^2 + 1} + \frac{Es + F}{(s^2 + 1)^2}$$

We can quickly find $A = -5/2$ and $B = 3/2$, then we'll plug these values into the decomposition, along with $s = 1$,

$$\frac{5(1^2 + 5(1) + 1)}{1(1 - 2)(1^2 + 1)^2} = -\frac{5}{2(1)} + \frac{3}{2(1 - 2)} + \frac{C(1) + D}{1^2 + 1} + \frac{E(1) + F}{(1^2 + 1)^2}$$

$$2C + 2D + E + F = -19$$

and then separately $s = -1$,

$$\frac{5((-1)^2 + 5(-1) + 1)}{(-1)((-1) - 2)((-1)^2 + 1)^2} = -\frac{5}{2(-1)} + \frac{3}{2((-1) - 2)} + \frac{C(-1) + D}{(-1)^2 + 1} + \frac{E(-1) + F}{((-1)^2 + 1)^2}$$

$$-2C + 2D - E + F = -13$$

and then separately $s = -2$,

$$\frac{5((-2)^2 + 5(-2) + 1)}{-2(-2 - 2)((-2)^2 + 1)^2} = -\frac{5}{2(-2)} + \frac{3}{2(-2 - 2)} + \frac{C(-2) + D}{(-2)^2 + 1} + \frac{E(-2) + F}{((-2)^2 + 1)^2}$$

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$$-10C + 5D - 2E + F = -25$$

and then separately $s = 3$,

$$\frac{5(3^2 + 5(3) + 1)}{3(3-2)(3^2+1)^2} = -\frac{5}{2(3)} + \frac{3}{2(3-2)} + \frac{C(3)+D}{3^2+1} + \frac{E(3)+F}{(3^2+1)^2}$$

$$30C + 10D + 3E + F = -25$$

to get the system of equations

$$2C + 2D + E + F = -19$$

$$-2C + 2D - E + F = -13$$

$$-10C + 5D - 2E + F = -25$$

$$30C + 10D + 3E + F = -25$$

We can use a matrix to solve the system for the values of C, D, E, and F.

$$\left[\begin{array}{cccc|c} 2 & 2 & 1 & 1 & -19 \\ -2 & 2 & -1 & 1 & -13 \\ -10 & 5 & -2 & 1 & -25 \\ 30 & 10 & 3 & 1 & -25 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{19}{2} \\ -2 & 2 & -1 & 1 & -13 \\ -10 & 5 & -2 & 1 & -25 \\ 30 & 10 & 3 & 1 & -25 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{19}{2} \\ 0 & 4 & 0 & 2 & -32 \\ -10 & 5 & -2 & 1 & -25 \\ 30 & 10 & 3 & 1 & -25 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{19}{2} \\ 0 & 4 & 0 & 2 & -32 \\ 0 & 15 & 3 & 6 & -120 \\ 30 & 10 & 3 & 1 & -25 \end{array}\right]$$

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$$\rightarrow \left[\begin{array}{cccc|c} 1 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{19}{2} \\ 0 & 4 & 0 & 2 & -32 \\ 0 & 15 & 3 & 6 & -120 \\ 0 & -20 & -12 & -14 & 260 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{19}{2} \\ 0 & 1 & 0 & \frac{1}{2} & -8 \\ 0 & 15 & 3 & 6 & -120 \\ 0 & -20 & -12 & -14 & 260 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 0 & \frac{1}{2} & 0 & -\frac{3}{2} \\ 0 & 1 & 0 & \frac{1}{2} & -8 \\ 0 & 15 & 3 & 6 & -120 \\ 0 & -20 & -12 & -14 & 260 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & \frac{1}{2} & 0 & -\frac{3}{2} \\ 0 & 1 & 0 & \frac{1}{2} & -8 \\ 0 & 0 & 3 & -\frac{3}{2} & 0 \\ 0 & -20 & -12 & -14 & 260 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 0 & \frac{1}{2} & 0 & -\frac{3}{2} \\ 0 & 1 & 0 & \frac{1}{2} & -8 \\ 0 & 0 & 3 & -\frac{3}{2} & 0 \\ 0 & 0 & -12 & -4 & 100 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & \frac{1}{2} & 0 & -\frac{3}{2} \\ 0 & 1 & 0 & \frac{1}{2} & -8 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & -12 & -4 & 100 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & \frac{1}{4} & -\frac{3}{2} \\ 0 & 1 & 0 & \frac{1}{2} & -8 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & -12 & -4 & 100 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & \frac{1}{4} & -\frac{3}{2} \\ 0 & 1 & 0 & \frac{1}{2} & -8 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & -10 & 100 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & \frac{1}{4} & -\frac{3}{2} \\ 0 & 1 & 0 & \frac{1}{2} & -8 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 & -10 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & \frac{1}{2} & -8 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 & -10 \end{array}\right]$$

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$$\rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & | & 1 \\ 0 & 1 & 0 & 0 & | & -3 \\ 0 & 0 & 1 & -\frac{1}{2} & | & 0 \\ 0 & 0 & 0 & 1 & | & -10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & | & 1 \\ 0 & 1 & 0 & 0 & | & -3 \\ 0 & 0 & 1 & 0 & | & -5 \\ 0 & 0 & 0 & 1 & | & -10 \end{bmatrix}$$

Solving this system gives $C = 1$, $D = -3$, $E = -5$, and $F = -10$. Then the Laplace transform is

$$Y(s) = -\frac{5}{2s} + \frac{3}{2(s-2)} + \frac{s-3}{s^2+1} - \frac{5s+10}{(s^2+1)^2}$$

$$Y(s) = -\frac{5}{2}\left(\frac{1}{s}\right) + \frac{3}{2}\left(\frac{1}{s-2}\right) + \frac{s}{s^2+1}$$

$$-3\left(\frac{1}{s^2+1}\right) - \frac{5}{2}\left(\frac{2s}{(s^2+1)^2}\right) - 5\left(\frac{2}{(s^2+1)^2}\right)$$

Applying the inverse Laplace transform to each term gives the general solution to the differential equation gives

$$y(t) = -\frac{5}{2} + \frac{3}{2}e^{2t} + \cos t - 3\sin t - \frac{5}{2}t\sin t - 5(\sin t - t\cos t)$$

$$y(t) = -\frac{5}{2} + \frac{3}{2}e^{2t} - 8\sin t + \cos t - \frac{5}{2}t\sin t + 5t\cos t$$

■ 5. Use the Laplace transform to solve the differential equation, given $y(0) = -2$, $y'(0) = 3$, and $y''(0) = 6$.

$$y''' - 6y'' + 12y' - 8y = 27e^{-t} - 2e^{t}$$

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Solution:

Apply the Laplace transform to both sides of the differential equation.

$$\mathscr{L}(y''') - 6\mathscr{L}(y'') + 12\mathscr{L}(y') - 8\mathscr{L}(y) = 27\mathscr{L}(e^{-t}) - 2\mathscr{L}(e^{t})$$

$$s^3Y(s) - s^2y(0) - sy'(0) - y''(0) - 6(s^2Y(s) - sy(0) - y'(0))$$

$$+12(sY(s) - y(0)) - 8Y(s) = \frac{27}{s+1} - \frac{2}{s-1}$$

Substitute the initial conditions $y(0) = -2$, $y'(0) = 3$, and $y''(0) = 6$,

$$Y(s)(s^3 - 6s^2 + 12s - 8) = \frac{27}{s+1} - \frac{2}{s-1} - 2s^2 + 15s - 36$$

$$Y(s)(s^3 - 6s^2 + 12s - 8) = \frac{-2s^4 + 15s^3 - 34s^2 + 10s + 7}{(s+1)(s-1)}$$

then solve for $Y(s)$.

$$Y(s) = \frac{-2s^4 + 15s^3 - 34s^2 + 10s + 7}{(s+1)(s-1)(s-2)^3}$$

Apply a partial fractions decomposition.

$$\frac{-2s^4 + 15s^3 - 34s^2 + 10s + 7}{(s+1)(s-1)(s-2)^3}$$

$$= \frac{A}{s+1} + \frac{B}{s-1} + \frac{C}{(s-2)^3} + \frac{D}{(s-2)^2} + \frac{E}{s-2}$$

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We can quickly find $A = -1$, $B = 2$, and $C = -7$, then we'll plug these values into the decomposition, along with $s = 0$,

$$\frac{-2(0)^4 + 15(0)^3 - 34(0)^2 + 10(0) + 7}{(0+1)(0-1)(0-2)^3}$$

$$= -\frac{1}{0+1} + \frac{2}{0-1} - \frac{7}{(0-2)^3} + \frac{D}{(0-2)^2} + \frac{E}{0-2}$$

$$D - 2E = 12$$

and then separately $s = -2$,

$$\frac{-2(-2)^4 + 15(-2)^3 - 34(-2)^2 + 10(-2) + 7}{(-2+1)(-2-1)(-2-2)^3}$$

$$= -\frac{1}{-2+1} + \frac{2}{-2-1} - \frac{7}{(-2-2)^3} + \frac{D}{(-2-2)^2} + \frac{E}{-2-2}$$

$$D - 4E = 18$$

to get the system of equations

$$D - 2E = 12$$

$$D - 4E = 18$$

Solving this system gives $D = 6$ and $E = -3$. Then the Laplace transform is

$$Y(s) = -\frac{1}{s+1} + \frac{2}{s-1} - \frac{7}{(s-2)^3} + \frac{6}{(s-2)^2} - \frac{3}{s-2}$$

Applying the inverse Laplace transform to each term gives the general solution to the differential equation gives

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$$y(t) = -e^{-t} + 2e^{t} - \frac{7}{2}t^2e^{2t} + 6te^{2t} - 3e^{2t}$$

■ 6. Use the Laplace transform to solve the differential equation, given $y(0) = 3$, $y'(0) = 0$, and $y''(0) = -9$.

$$y''' + 3y'' + 4y' + 2y = -8\sin t - 6\cos t$$

Solution:

Apply the Laplace transform to both sides of the differential equation.

$$\mathcal{L}(y''') + 3\mathcal{L}(y'') + 4\mathcal{L}(y') + 2\mathcal{L}(y) = -8\mathcal{L}(\sin t) - 6\mathcal{L}(\cos t)$$

$$s^3Y(s) - s^2y(0) - sy'(0) - y''(0) + 3s^2Y(s) - 3sy(0) - 3y'(0)$$

$$+4sY(s) - 4y(0) + 2Y(s) = -\frac{8}{s^2+1} - \frac{6s}{s^2+1}$$

Substitute the initial conditions $y(0) = 3$, $y'(0) = 0$, and $y''(0) = -9$,

$$Y(s)(s^3 + 3s^2 + 4s + 2) = \frac{-8 - 6s}{s^2+1} + 3s^2 + 9s + 3$$

$$Y(s)(s^3 + 3s^2 + 4s + 2) = \frac{3s^4 + 9s^3 + 6s^2 + 3s - 5}{s^2+1}$$

then solve for $Y(s)$.

$$Y(s) = \frac{3s^4 + 9s^3 + 6s^2 + 3s - 5}{(s^2+1)(s^2+2s+2)(s+1)}$$

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Apply a partial fractions decomposition.

$$\frac{3s^4+9s^3+6s^2+3s-5}{(s^2+1)(s^2+2s+2)(s+1)}=\frac{A}{s+1}+\frac{Bs+C}{s^2+1}+\frac{Ds+E}{s^2+2s+2}$$

We can quickly find $A=-4$, then we'll plug this value into the decomposition, along with $s=-2$,

$$\frac{3(-2)^4+9(-2)^3+6(-2)^2+3(-2)-5}{((-2)^2+1)((-2)^2+2(-2)+2)(-2+1)}$$

$$=-\frac{4}{-2+1}+\frac{B(-2)+C}{(-2)^2+1}+\frac{D(-2)+E}{(-2)^2+2(-2)+2}$$

$$4B-2C+10D-5E=29$$

and then separately $s=0$,

$$\frac{3(0)^4+9(0)^3+6(0)^2+3(0)-5}{(0^2+1)(0^2+2(0)+2)(0+1)}=-\frac{4}{0+1}+\frac{B(0)+C}{0^2+1}+\frac{D(0)+E}{0^2+2(0)+2}$$

$$2C+E=3$$

and then separately $s=1$,

$$\frac{3(1)^4+9(1)^3+6(1)^2+3(1)-5}{(1^2+1)(1^2+2(1)+2)(1+1)}=-\frac{4}{1+1}+\frac{B(1)+C}{1^2+1}+\frac{D(1)+E}{1^2+2(1)+2}$$

$$5B+5C+2D+2E=28$$

and then separately $s=2$,

$$\frac{3(2)^4+9(2)^3+6(2)^2+3(2)-5}{(2^2+1)(2^2+2(2)+2)(2+1)}=-\frac{4}{2+1}+\frac{B(2)+C}{2^2+1}+\frac{D(2)+E}{2^2+2(2)+2}$$

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$$4B + 2C + 2D + E = 23$$

to get the system of equations

$$4B - 2C + 10D - 5E = 29$$

$$2C + E = 3$$

$$5B + 5C + 2D + 2E = 28$$

$$4B + 2C + 2D + E = 23$$

We can use a matrix to solve the system for the values of B, C, D, and E.

$$\begin{bmatrix} 4 & -2 & 10 & -5 & | & 29 \\ 0 & 2 & 0 & 1 & | & 3 \\ 5 & 5 & 2 & 2 & | & 28 \\ 4 & 2 & 2 & 1 & | & 23 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & \frac{5}{2} & -\frac{5}{4} & | & \frac{29}{4} \\ 0 & 2 & 0 & 1 & | & 3 \\ 5 & 5 & 2 & 2 & | & 28 \\ 4 & 2 & 2 & 1 & | & 23 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & \frac{5}{2} & -\frac{5}{4} & | & \frac{29}{4} \\ 0 & 2 & 0 & 1 & | & 3 \\ 0 & \frac{15}{2} & -\frac{21}{2} & \frac{33}{4} & | & -\frac{33}{4} \\ 4 & 2 & 2 & 1 & | & 23 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & \frac{5}{2} & -\frac{5}{4} & | & \frac{29}{4} \\ 0 & 2 & 0 & 1 & | & 3 \\ 0 & \frac{15}{2} & -\frac{21}{2} & \frac{33}{4} & | & -\frac{33}{4} \\ 0 & 4 & -8 & 6 & | & -6 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & \frac{5}{2} & -\frac{5}{4} & | & \frac{29}{4} \\ 0 & 1 & 0 & \frac{1}{2} & | & \frac{3}{2} \\ 0 & \frac{15}{2} & -\frac{21}{2} & \frac{33}{4} & | & -\frac{33}{4} \\ 0 & 4 & -8 & 6 & | & -6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & \frac{5}{2} & -1 & | & 8 \\ 0 & 1 & 0 & \frac{1}{2} & | & \frac{3}{2} \\ 0 & \frac{15}{2} & -\frac{21}{2} & \frac{33}{4} & | & -\frac{33}{4} \\ 0 & 4 & -8 & 6 & | & -6 \end{bmatrix}$$

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$$\rightarrow \left[\begin{array}{cccc|c} 1 & 0 & \frac{5}{2} & -1 & 8 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{3}{2} \\ 0 & 0 & -\frac{21}{2} & \frac{9}{2} & -\frac{39}{2} \\ 0 & 4 & -8 & 6 & -6 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & \frac{5}{2} & -1 & 8 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{3}{2} \\ 0 & 0 & 1 & -\frac{3}{7} & \frac{13}{7} \\ 0 & 4 & -8 & 6 & -6 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 0 & \frac{5}{2} & -1 & 8 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{3}{2} \\ 0 & 0 & 1 & -\frac{3}{7} & \frac{13}{7} \\ 0 & 0 & -8 & 4 & -12 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & \frac{1}{14} & \frac{47}{14} \\ 0 & 1 & 0 & \frac{1}{2} & \frac{3}{2} \\ 0 & 0 & 1 & -\frac{3}{7} & \frac{13}{7} \\ 0 & 0 & -8 & 4 & -12 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & \frac{1}{14} & \frac{47}{14} \\ 0 & 1 & 0 & \frac{1}{2} & \frac{3}{2} \\ 0 & 0 & 1 & -\frac{3}{7} & \frac{13}{7} \\ 0 & 0 & 0 & \frac{4}{7} & \frac{20}{7} \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & \frac{1}{14} & \frac{47}{14} \\ 0 & 1 & 0 & \frac{1}{2} & \frac{3}{2} \\ 0 & 0 & 1 & -\frac{3}{7} & \frac{13}{7} \\ 0 & 0 & 0 & 1 & 5 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{3}{2} \\ 0 & 0 & 1 & -\frac{3}{7} & \frac{13}{7} \\ 0 & 0 & 0 & 1 & 5 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & -\frac{3}{7} & \frac{13}{7} \\ 0 & 0 & 0 & 1 & 5 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 1 & 5 \end{array}\right]$$

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Solving this system gives $B = 9/4$, $C = -5/2$, $D = 11/2$, and $E = 8$. Then the Laplace transform is

$$Y(s) = -\frac{4}{s+1} + \frac{3s-1}{s^2+1} + \frac{4s+5}{s^2+2s+2}$$

$$Y(s) = -4\left(\frac{1}{s-(-1)}\right) + 3\left(\frac{s}{s^2+1^2}\right) - \frac{1}{s^2+1^2}$$

$$+4\left(\frac{s-(-1)}{(s-(-1))^2+1^2}\right) - 4\left(\frac{1}{(s-(-1))^2+1^2}\right)$$

$$+5\left(\frac{1}{(s-(-1))^2+1^2}\right)$$

Applying the inverse Laplace transform to each term gives the general solution to the differential equation gives

$$y(t) = -4e^{-t} + 3\cos t - \sin t + 4e^{-t}\cos t - 4e^{-t}\sin t + 5e^{-t}\sin t$$

$$y(t) = -4e^{-t} + 3\cos t - \sin t + 4e^{-t}\cos t + e^{-t}\sin t$$

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SYSTEMS OF HIGHER ORDER EQUATIONS

■ 1. Solve the system of differential equations.

$$x_1' = 2x_1$$

$$x_2' = 4x_3 - x_2$$

$$x_3' = -3x_1$$

Solution:

We'll need to start by finding the matrix $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 4 \\ -3 & 0 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 4 \\ -3 & 0 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2-\lambda & 0 & 0 \\ 0 & -1-\lambda & 4 \\ -3 & 0 & -\lambda \end{bmatrix}$$

and then finding its determinant $|A - \lambda I|$.

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & 0 & 0 \\ 0 & -1-\lambda & 4 \\ -3 & 0 & -\lambda \end{vmatrix}$$

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$$|A-\lambda I| = (2-\lambda)\begin{vmatrix} -1-\lambda & 4 \\ 0 & -\lambda \end{vmatrix} - (0)\begin{vmatrix} 0 & 4 \\ -3 & -\lambda \end{vmatrix} + (0)\begin{vmatrix} 0 & -1-\lambda \\ -3 & 0 \end{vmatrix}$$

$$|A-\lambda I| = (2-\lambda)((-1-\lambda)(-\lambda)-(4)(0))-0+0$$

$$|A-\lambda I| = (2-\lambda)(-1-\lambda)(-\lambda)$$

Solve the characteristic equation for the Eigenvalues.

$$(2-\lambda)(-1-\lambda)(-\lambda) = 0$$

$$\lambda = -1,\ 0,\ 2$$

For the Eigenvalue $\lambda_1 = -1$, we find

$$A+1I = \begin{bmatrix} 2+1 & 0 & 0 \\ 0 & -1+1 & 4 \\ -3 & 0 & 1 \end{bmatrix}$$

$$A+1I = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 0 & 4 \\ -3 & 0 & 1 \end{bmatrix}$$

Rewriting this matrix as a system of equations,

$$3k_1 = 0$$

$$4k_3 = 0$$

$$-3k_1 + k_3 = 0$$

we get $k_1 = 0$, $k_3 = 0$, and we'll choose $k_2 = 1$. Then the solution vector is

$$\overrightarrow{x_1} = \overrightarrow{k_1}e^{\lambda_1 t}$$

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$$\overrightarrow{x_1} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{-t}$$

For the Eigenvalue $\lambda_2 = 0$, we find

$$A - (0)I = \begin{bmatrix} 2-0 & 0 & 0 \\ 0 & -1-0 & 4 \\ -3 & 0 & 0 \end{bmatrix}$$

$$A - (0)I = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 4 \\ -3 & 0 & 0 \end{bmatrix}$$

Rewriting this matrix as a system of equations,

$$2k_1 = 0$$

$$-k_2 + 4k_3 = 0$$

$$-3k_1 = 0$$

we get $k_1 = 0$, and we'll choose $k_3 = 1$ to get $k_2 = 4$. The the solution vector is

$$\overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix} e^{0t} = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}$$

For the Eigenvalue $\lambda_3 = 2$, we find

$$A - 2I = \begin{bmatrix} 2-2 & 0 & 0 \\ 0 & -1-2 & 4 \\ -3 & 0 & -2 \end{bmatrix}$$

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$$A - 2I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -3 & 4 \\ -3 & 0 & -2 \end{bmatrix}$$

Rewriting this matrix as a system of equations,

$$-3k_2 + 4k_3 = 0$$

$$-3k_1 - 2k_3 = 0$$

we'll choose $k_3 = 3$ to get $k_1 = -2$ and $k_2 = 4$. Then the solution vector is

$$\vec{x_3} = \vec{k_3} e^{\lambda_3 t}$$

$$\vec{x_3} = \begin{bmatrix} -2 \\ 4 \\ 3 \end{bmatrix} e^{2t}$$

So the general solution to the homogeneous system is

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2} + c_3\vec{x_3}$$

$$\vec{x} = c_1 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} -2 \\ 4 \\ 3 \end{bmatrix} e^{2t}$$

2. Solve the system of differential equations.

$$x_1' = 2x_2 - 2x_3$$

$$x_2' = 2x_1 + 4x_2 + 4x_3$$

$$x_3' = -2x_1 + 4x_2 - 3x_3$$

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Solution:

We'll need to start by finding the matrix $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 0 & 2 & -2 \\ 2 & 4 & 4 \\ -2 & 4 & -3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 0 & 2 & -2 \\ 2 & 4 & 4 \\ -2 & 4 & -3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} -\lambda & 2 & -2 \\ 2 & 4 - \lambda & 4 \\ -2 & 4 & -3 - \lambda \end{bmatrix}$$

and then finding its determinant $|A - \lambda I|$.

$$|A - \lambda I| = \begin{vmatrix} -\lambda & 2 & -2 \\ 2 & 4 - \lambda & 4 \\ -2 & 4 & -3 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (-\lambda)\begin{vmatrix} 4 - \lambda & 4 \\ 4 & -3 - \lambda \end{vmatrix} - 2\begin{vmatrix} 2 & 4 \\ -2 & -3 - \lambda \end{vmatrix} + (-2)\begin{vmatrix} 2 & 4 - \lambda \\ -2 & 4 \end{vmatrix}$$

$$|A - \lambda I| = (-\lambda)((4 - \lambda)(-3 - \lambda) - (4)(4)) - 2((2)(-3 - \lambda) - (-2)(4))$$

$$-2((2)(4) - (4 - \lambda)(-2))$$

$$|A - \lambda I| = (-\lambda)(\lambda^2 - \lambda - 28) - 2(-2\lambda + 2) - 2(16 - 2\lambda)$$

$$|A - \lambda I| = -\lambda^3 + \lambda^2 + 36\lambda - 36$$

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$$|A - \lambda I| = (\lambda^2 - 36)(1 - \lambda)$$

Solve the characteristic equation for the Eigenvalues.

$$(\lambda - 6)(\lambda + 6)(1 - \lambda) = 0$$

$$\lambda = -6,\ 1,\ 6$$

For the Eigenvalue $\lambda_1 = -6$, we find

$$A + 6I = \begin{bmatrix} 6 & 2 & -2 \\ 2 & 4+6 & 4 \\ -2 & 4 & -3+6 \end{bmatrix}$$

$$A + 6I = \begin{bmatrix} 6 & 2 & -2 \\ 2 & 10 & 4 \\ -2 & 4 & 3 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 2 & 10 & 4 \\ 6 & 2 & -2 \\ -2 & 4 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 5 & 2 \\ 6 & 2 & -2 \\ -2 & 4 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 5 & 2 \\ 0 & -28 & -14 \\ 0 & 14 & 7 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 5 & 2 \\ 0 & 14 & 7 \\ 0 & -28 & -14 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 5 & 2 \\ 0 & 14 & 7 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 5 & 2 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}$$

Rewriting this matrix as a system of equations,

$$k_1 - \frac{1}{2}k_3 = 0$$

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$$k_2 + \frac{1}{2}k_3 = 0$$

we'll choose $k_3 = 2$ to get $k_1 = 1$ and $k_2 = -1$. Then the solution vector is

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} e^{-6t}$$

For the Eigenvalue $\lambda_2 = 1$, we find

$$A - 1I = \begin{bmatrix} -1 & 2 & -2 \\ 2 & 4-1 & 4 \\ -2 & 4 & -3-1 \end{bmatrix}$$

$$A - 1I = \begin{bmatrix} -1 & 2 & -2 \\ 2 & 3 & 4 \\ -2 & 4 & -4 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & -2 & 2 \\ 2 & 3 & 4 \\ -2 & 4 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 2 \\ 0 & 7 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Rewriting this matrix as a system of equations,

$$k_1 + 2k_3 = 0$$

$$k_2 = 0$$

we get $k_2 = 0$ and we'll choose $k_3 = 1$ to get $k_1 = -2$. Then the solution vector is

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$$\overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} e^t$$

For the Eigenvalue $\lambda_3 = 6$, we find

$$A - 6I = \begin{bmatrix} -6 & 2 & -2 \\ 2 & 4-6 & 4 \\ -2 & 4 & -3-6 \end{bmatrix}$$

$$A - 6I = \begin{bmatrix} -6 & 2 & -2 \\ 2 & -2 & 4 \\ -2 & 4 & -9 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 2 & -2 & 4 \\ -6 & 2 & -2 \\ -2 & 4 & -9 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 2 \\ 0 & -4 & 10 \\ 0 & 2 & -5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 2 \\ 0 & -4 & 10 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -\frac{5}{2} \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & -\frac{5}{2} \\ 0 & 0 & 0 \end{bmatrix}$$

Rewriting this matrix as a system of equations,

$$k_1 - \frac{1}{2}k_3 = 0$$

$$k_2 - \frac{5}{2}k_3 = 0$$

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we'll choose $k_3 = 2$ to get $k_1 = 1$ and $k_2 = 5$. Then the solution vector is

$$\vec{x_3} = \vec{k_3} e^{\lambda_3 t}$$

$$\vec{x_3} = \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} e^{6t}$$

So the general solution to the homogeneous system is

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2} + c_3\vec{x_3}$$

$$\vec{x} = c_1 \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} e^{-6t} + c_2 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} e^{t} + c_3 \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} e^{6t}$$

3. Solve the system of differential equations.

$$x_1' = 2x_1 + 4x_3$$

$$x_2' = -x_1 + 2x_2$$

$$x_3' = -x_3$$

Solution:

We'll need to start by finding the matrix $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 2 & 0 & 4 \\ -1 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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$$A - \lambda I = \begin{bmatrix} 2 & 0 & 4 \\ -1 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2-\lambda & 0 & 4 \\ -1 & 2-\lambda & 0 \\ 0 & 0 & -1-\lambda \end{bmatrix}$$

and then finding its determinant $|A - \lambda I|$.

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & 0 & 4 \\ -1 & 2-\lambda & 0 \\ 0 & 0 & -1-\lambda \end{vmatrix}$$

$$|A - \lambda I| = (2-\lambda)\begin{vmatrix} 2-\lambda & 0 \\ 0 & -1-\lambda \end{vmatrix} - (0)\begin{vmatrix} -1 & 0 \\ 0 & -1-\lambda \end{vmatrix} + 4\begin{vmatrix} -1 & 2-\lambda \\ 0 & 0 \end{vmatrix}$$

$$|A - \lambda I| = (2-\lambda)((2-\lambda)(-1-\lambda) - (0)(0)) - 0 + 4((-1)(0) - (2-\lambda)(0))$$

$$|A - \lambda I| = (2-\lambda)(2-\lambda)(-1-\lambda)$$

Solve the characteristic equation for the Eigenvalues.

$$(2-\lambda)(2-\lambda)(-1-\lambda) = 0$$

$$\lambda = -1, 2, 2$$

For the Eigenvalue $\lambda_1 = -1$, we find

$$A + 1I = \begin{bmatrix} 2+1 & 0 & 4 \\ -1 & 2+1 & 0 \\ 0 & 0 & -1+1 \end{bmatrix}$$

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$$A + 1I = \begin{bmatrix} 3 & 0 & 4 \\ -1 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Rewriting this matrix as a system of equations,

$$3k_1 + 4k_3 = 0$$

$$-k_1 + 3k_2 = 0$$

we'll choose $k_3 = 9$ to get $k_1 = -12$ and $k_2 = -4$. Then the solution vector is

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} -12 \\ -4 \\ 9 \end{bmatrix} e^{-t}$$

For the Eigenvalue $\lambda_2 = \lambda_3 = 2$, we find

$$A - 2I = \begin{bmatrix} 2-2 & 0 & 4 \\ -1 & 2-2 & 0 \\ 0 & 0 & -1-2 \end{bmatrix}$$

$$A - 2I = \begin{bmatrix} 0 & 0 & 4 \\ -1 & 0 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

Rewriting this matrix as a system of equations,

$$4k_3 = 0$$

$$-k_1 = 0$$

$$-3k_3 = 0$$

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we get $k_1 = 0$ and $k_3 = 0$, and we'll choose $k_2 = 1$. Then the solution vector is

$$\vec{x_2} = \vec{k_2} e^{\lambda_2 t}$$

$$\vec{x_2} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{2t}$$

Because we only find one Eigenvector for the two Eigenvalues $\lambda_2 = \lambda_3 = 2$, we have to use $\vec{k_2} = (0,1,0)$ to find a second solution.

$$(A - \lambda_2 I)\vec{p_1} = \vec{k_2}$$

$$\begin{bmatrix} 0 & 0 & 4 \\ -1 & 0 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Rewriting this matrix as a system of equations,

$$4p_3 = 0$$

$$-p_1 = 1$$

$$-3p_3 = 0$$

we get $p_1 = -1$ and $p_3 = 0$, and we'll choose $p_2 = 0$. Then the second solution will be

$$\vec{x_3} = \vec{k_2} t e^{\lambda_2 t} + \vec{p_1} e^{\lambda_2 t}$$

$$\vec{x_3} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} t e^{2t} + \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} e^{2t}$$

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So the general solution to the homogeneous system is

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2} + c_3\vec{x_3}$$

$$\vec{x} = c_1\begin{bmatrix}-12\\-4\\9\end{bmatrix}e^{-t} + c_2\begin{bmatrix}0\\1\\0\end{bmatrix}e^{2t} + c_3\left(\begin{bmatrix}0\\1\\0\end{bmatrix}te^{2t} + \begin{bmatrix}-1\\0\\0\end{bmatrix}e^{2t}\right)$$

■ 4. Solve the system of differential equations.

$$x_2' = -x_1 + 3x_2 + 5x_3$$

$$x_3' = 2x_1 - 2x_2 + x_3$$

Solution:

We'll need to start by finding the matrix $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix}0 & 0 & 0\\-1 & 3 & 5\\2 & -2 & 1\end{bmatrix} - \lambda\begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix}0 & 0 & 0\\-1 & 3 & 5\\2 & -2 & 1\end{bmatrix} - \begin{bmatrix}\lambda & 0 & 0\\0 & \lambda & 0\\0 & 0 & \lambda\end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix}-\lambda & 0 & 0\\-1 & 3-\lambda & 5\\2 & -2 & 1-\lambda\end{bmatrix}$$

and then finding its determinant $|A - \lambda I|$.

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$$|A-\lambda I| = \begin{vmatrix} -\lambda & 0 & 0 \\ -1 & 3-\lambda & 5 \\ 2 & -2 & 1-\lambda \end{vmatrix}$$

$$|A-\lambda I| = (-\lambda)\begin{vmatrix} 3-\lambda & 5 \\ -2 & 1-\lambda \end{vmatrix} - (0)\begin{vmatrix} -1 & 5 \\ 2 & 1-\lambda \end{vmatrix} + (0)\begin{vmatrix} -1 & 3-\lambda \\ 2 & -2 \end{vmatrix}$$

$$|A-\lambda I| = (-\lambda)((3-\lambda)(1-\lambda)-(5)(-2)) - 0 + 0$$

$$|A-\lambda I| = (-\lambda)(\lambda^2 - 4\lambda + 13)$$

Solve the characteristic equation for the Eigenvalues.

$$(-\lambda)(\lambda^2 - 4\lambda + 13) = 0$$

$$\lambda = 0,\ 2 \pm 3i$$

For the Eigenvalue $\lambda_1 = 0$, we find

$$A-(0)I = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 3-0 & 5 \\ 2 & -2 & 1-0 \end{bmatrix}$$

$$A-(0)I = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 3 & 5 \\ 2 & -2 & 1 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 1 & -3 & -5 \\ 2 & -2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 & -5 \\ 0 & 4 & 11 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 & -5 \\ 0 & 1 & \frac{11}{4} \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & \frac{13}{4} \\ 0 & 1 & \frac{11}{4} \\ 0 & 0 & 0 \end{bmatrix}$$

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Rewriting this matrix as a system of equations,

$$k_1 + \frac{13}{4}k_3 = 0$$

$$k_2 + \frac{11}{4}k_3 = 0$$

we'll choose $k_3 = 4$ to get $k_1 = -13$ and $k_2 = -11$. Then the solution vector is

$$\overrightarrow{x_1} = \overrightarrow{k_1}e^{\lambda_1 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} -13 \\ -11 \\ 4 \end{bmatrix}$$

For the Eigenvalue $\lambda_2 = 2 + 3i$, we find

$$A - (2+3i)I = \begin{bmatrix} -2-3i & 0 & 0 \\ -1 & 3-2-3i & 5 \\ 2 & -2 & 1-2-3i \end{bmatrix}$$

$$A - (2+3i)I = \begin{bmatrix} -2-3i & 0 & 0 \\ -1 & 1-3i & 5 \\ 2 & -2 & -1-3i \end{bmatrix}$$

Rewriting this matrix as a system of equations,

$$(-2-3i)k_1 = 0$$

$$-k_1 + (1-3i)k_2 + 5k_3 = 0$$

$$2k_1 - 2k_2 + (-1-3i)k_3 = 0$$

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we get $k_1 = 0$ and we'll choose $k_3 = -2$ to get $k_2 = 1 + 3i$. Then the solution vector is

$$\vec{x_2} = \vec{k_2} e^{\lambda_2 t}$$

$$\vec{x_2} = \begin{bmatrix} 0 \\ 1+3i \\ -2 \end{bmatrix} e^{(2+3i)t}$$

For the Eigenvalue $\lambda_3 = 2 - 3i$, we find

$$A - (2-3i)I = \begin{bmatrix} -2+3i & 0 & 0 \\ -1 & 3-2+3i & 5 \\ 2 & -2 & 1-2+3i \end{bmatrix}$$

$$A - (2-3i)I = \begin{bmatrix} -2+3i & 0 & 0 \\ -1 & 1+3i & 5 \\ 2 & -2 & -1+3i \end{bmatrix}$$

Rewriting this matrix as a system of equations,

$$(-2+3i)k_1 = 0$$

$$-k_1 + (1+3i)k_2 + 5k_3 = 0$$

$$2k_1 - 2k_2 + (-1+3i)k_3 = 0$$

we get $k_1 = 0$ and we'll choose $k_3 = -2$ to get $k_2 = 1 - 3i$. Then the solution vector is

$$\vec{x_3} = \vec{k_3} e^{\lambda_3 t}$$

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$$\overrightarrow{x_3} = \begin{bmatrix} 0 \\ 1-3i \\ -2 \end{bmatrix} e^{(2-3i)t}$$

So the general solution to the homogeneous system is

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} -13 \\ -11 \\ 4 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1+3i \\ -2 \end{bmatrix} e^{(2+3i)t} + c_3 \begin{bmatrix} 0 \\ 1-3i \\ -2 \end{bmatrix} e^{(2-3i)t}$$

5. Solve the system of differential equations.

$$x_1' = x_1 - x_2$$

$$x_2' = 5x_1 - x_2$$

$$x_3' = -3x_1 + x_2 + 4x_3$$

Solution:

We'll need to start by finding the matrix $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} 1 & -1 & 0 \\ 5 & -1 & 0 \\ -3 & 1 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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$$A - \lambda I = \begin{bmatrix} 1 & -1 & 0 \\ 5 & -1 & 0 \\ -3 & 1 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & -1 & 0 \\ 5 & -1-\lambda & 0 \\ -3 & 1 & 4-\lambda \end{bmatrix}$$

and then finding its determinant $|A - \lambda I|$.

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & -1 & 0 \\ 5 & -1-\lambda & 0 \\ -3 & 1 & 4-\lambda \end{vmatrix}$$

$$|A - \lambda I| = (1-\lambda)\begin{vmatrix} -1-\lambda & 0 \\ 1 & 4-\lambda \end{vmatrix} - (-1)\begin{vmatrix} 5 & 0 \\ -3 & 4-\lambda \end{vmatrix} + (0)\begin{vmatrix} 5 & -1-\lambda \\ -3 & 1 \end{vmatrix}$$

$$|A - \lambda I| = (1-\lambda)((-1-\lambda)(4-\lambda) - (1)(0)) + ((5)(4-\lambda) - (-3)(0)) + 0$$

$$|A - \lambda I| = (1-\lambda)(-1-\lambda)(4-\lambda) + (5)(4-\lambda)$$

$$|A - \lambda I| = (4-\lambda)(\lambda^2 + 4)$$

Solve the characteristic equation for the Eigenvalues.

$$(4-\lambda)(\lambda^2+4) = 0$$

$$\lambda = 4, \ \pm 2i$$

For the Eigenvalue $\lambda_1 = 4$, we find

$$A - 4I = \begin{bmatrix} 1-4 & -1 & 0 \\ 5 & -1-4 & 0 \\ -3 & 1 & 4-4 \end{bmatrix}$$

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$$A - 4I = \begin{bmatrix} -3 & -1 & 0 \\ 5 & -5 & 0 \\ -3 & 1 & 0 \end{bmatrix}$$

Put this matrix into reduced row-echelon form.

$$\begin{bmatrix} 5 & -5 & 0 \\ -3 & -1 & 0 \\ -3 & 1 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 0 \\ -3 & -1 & 0 \\ -3 & 1 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 0 \\ 0 & -4 & 0 \\ 0 & -2 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Rewriting this matrix as a system of equations,

$$k_1 = 0$$

$$k_2 = 0$$

we get $k_1 = 0$ and $k_2 = 0$ and we'll choose $k_3 = 1$. Then the solution vector is

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{4t}$$

For the Eigenvalue $\lambda_2 = 2i$, we find

$$A - 2iI = \begin{bmatrix} 1 - 2i & -1 & 0 \\ 5 & -1 - 2i & 0 \\ -3 & 1 & 4 - 2i \end{bmatrix}$$

Rewriting this matrix as a system of equations,

$$(1 - 2i)k_1 - k_2 = 0$$

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$$5k_1 + (-1 - 2i)k_2 = 0$$

$$-3k_1 + k_2 + (4 - 2i)k_3 = 0$$

we'll choose $k_3 = 2$ to get $k_1 = 1 - 3i$ and $k_2 = -5 - 5i$. Then the solution vector is

$$\overrightarrow{x_2} = \overrightarrow{k_2}e^{\lambda_2 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 1 - 3i \\ -5 - 5i \\ 2 \end{bmatrix} e^{2it}$$

For the Eigenvalue $\lambda_3 = -2i$, we find

$$A + 2iI = \begin{bmatrix} 1 + 2i & -1 & 0 \\ 5 & -1 + 2i & 0 \\ -3 & 1 & 4 + 2i \end{bmatrix}$$

Rewriting this matrix as a system of equations,

$$(1 + 2i)k_1 - k_2 = 0$$

$$5k_1 + (-1 + 2i)k_2 = 0$$

$$-3k_1 + k_2 + (4 + 2i)k_3 = 0$$

we'll choose $k_3 = 2$ to get $k_1 = 1 + 3i$ and $k_2 = -5 + 5i$. Then the solution vector is

$$\overrightarrow{x_3} = \overrightarrow{k_3}e^{\lambda_3 t}$$

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$$\vec{x_3} = \begin{bmatrix} 1+3i \\ -5+5i \\ 2 \end{bmatrix} e^{-2it}$$

So the general solution to the homogeneous system is

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2} + c_3\vec{x_3}$$

$$\vec{x} = c_1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} 1-3i \\ -5-5i \\ 2 \end{bmatrix} e^{2it} + c_3 \begin{bmatrix} 1+3i \\ -5+5i \\ 2 \end{bmatrix} e^{-2it}$$

6. Solve the system of differential equations.

$$x_1' = -x_1 + 3x_2 - x_3$$

$$x_2' = -x_2 + 2x_3$$

$$x_3' = -x_3$$

Solution:

We'll need to start by finding the matrix $A - \lambda I$,

$$A - \lambda I = \begin{bmatrix} -1 & 3 & -1 \\ 0 & -1 & 2 \\ 0 & 0 & -1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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$$A-\lambda I=\begin{bmatrix}-1 & 3 & -1\\ 0 & -1 & 2\\ 0 & 0 & -1\end{bmatrix}-\begin{bmatrix}\lambda & 0 & 0\\ 0 & \lambda & 0\\ 0 & 0 & \lambda\end{bmatrix}$$

$$A-\lambda I=\begin{bmatrix}-1-\lambda & 3 & -1\\ 0 & -1-\lambda & 2\\ 0 & 0 & -1-\lambda\end{bmatrix}$$

and then finding its determinant $|A-\lambda I|$.

$$|A-\lambda I|=\begin{vmatrix}-1-\lambda & 3 & -1\\ 0 & -1-\lambda & 2\\ 0 & 0 & -1-\lambda\end{vmatrix}$$

$$|A-\lambda I|=(-1-\lambda)\begin{vmatrix}-1-\lambda & 2\\ 0 & -1-\lambda\end{vmatrix}-3\begin{vmatrix}0 & 2\\ 0 & -1-\lambda\end{vmatrix}+(-1)\begin{vmatrix}0 & -1-\lambda\\ 0 & 0\end{vmatrix}$$

$$|A-\lambda I|=(-1-\lambda)((-1-\lambda)(-1-\lambda)-(2)(0))-0+0$$

$$|A-\lambda I|=(-1-\lambda)(-1-\lambda)(-1-\lambda)$$

Solve the characteristic equation for the Eigenvalues.

$$(-1-\lambda)(-1-\lambda)(-1-\lambda)=0$$

$$\lambda=-1,\ -1,\ -1$$

For the Eigenvalue $\lambda_1=\lambda_2=\lambda_3=-1$, we find

$$A+1I=\begin{bmatrix}-1+1 & 3 & -1\\ 0 & -1+1 & 2\\ 0 & 0 & -1+1\end{bmatrix}$$

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$$A + 1I = \begin{bmatrix} 0 & 3 & -1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Rewriting this matrix as a system of equations,

$$3k_2 - k_3 = 0$$

$$2k_3 = 0$$

we get $k_2 = 0$ and $k_3 = 0$, and we'll choose $k_1 = 1$. Then the solution vector is

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-t}$$

Because we only find one Eigenvector for the three Eigenvalues $\lambda_1 = \lambda_2 = \lambda_3 = -1$, we have to use $\overrightarrow{k_1} = (1,0,0)$ to find a second solution.

$$(A - \lambda_2 I)\overrightarrow{p_1} = \overrightarrow{k_1}$$

$$\begin{bmatrix} 0 & 3 & -1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Rewriting this matrix as a system of equations,

$$3p_2 - p_3 = 1$$

$$2p_3 = 0$$

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we get $p_3 = 0$ and $p_2 = 1/3$, and we'll choose $p_1 = 1$. Then the second solution will be

$$\vec{x_2} = \vec{k_1} t e^{\lambda_2 t} + \vec{p_1} e^{\lambda_2 t}$$

$$\vec{x_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} te^{-t} + \begin{bmatrix} 1 \\ \frac{1}{3} \\ 0 \end{bmatrix} e^{-t}$$

Now we'll use $\vec{p_1} = (1,1/3,0)$ to find a third solution.

$$(A - \lambda_3 I)\vec{q_1} = \vec{p_1}$$

$$\begin{bmatrix} 0 & 3 & -1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{1}{3} \\ 0 \end{bmatrix}$$

Rewriting this matrix as a system of equations,

$$3q_2 - q_3 = 1$$

$$2q_3 = \frac{1}{3}$$

we get $q_2 = 7/18$ and $q_3 = 1/6$, and we'll choose $q_1 = 1$. Then the second solution will be

$$\vec{x_3} = \vec{k_1} \frac{t^2}{2} e^{\lambda_3 t} + \vec{p_1} t e^{\lambda_3 t} + \vec{q_1} e^{\lambda_3 t}$$

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$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \frac{t^2}{2} e^{-t} + \begin{bmatrix} 1 \\ \frac{1}{3} \\ 0 \end{bmatrix} te^{-t} + \begin{bmatrix} 1 \\ \frac{7}{18} \\ \frac{1}{6} \end{bmatrix} e^{-t}$$

So the general solution to the homogeneous system is

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-t} + c_2 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} te^{-t} + \begin{bmatrix} 1 \\ \frac{1}{3} \\ 0 \end{bmatrix} e^{-t} \right) + c_3 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \frac{t^2}{2} e^{-t} + \begin{bmatrix} 1 \\ \frac{1}{3} \\ 0 \end{bmatrix} te^{-t} + \begin{bmatrix} 1 \\ \frac{7}{18} \\ \frac{1}{6} \end{bmatrix} e^{-t} \right)$$

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SERIES SOLUTIONS OF HIGHER ORDER EQUATIONS

■ 1. Find a power series solution in x to the differential equation.

$$y''' - xy' = 0$$

Solution:

Substitute y' and y''' into the differential equation.

$$\sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-3} - x\sum_{n=1}^{\infty} c_n n x^{n-1} = 0$$

$$\sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-3} - \sum_{n=1}^{\infty} c_n n x^{n} = 0$$

To put these series in phase, we'll pull the x^0 term out of the first series.

$$c_3 \cdot 3(3-1)(3-2)x^{3-3} + \sum_{n=4}^{\infty} c_n n(n-1)(n-2)x^{n-3} - \sum_{n=1}^{\infty} c_n n x^{n} = 0$$

$$6c_3 + \sum_{n=4}^{\infty} c_n n(n-1)(n-2)x^{n-3} - \sum_{n=1}^{\infty} c_n n x^{n} = 0$$

Now the series are in phase, but the indices don't match. We can substitute $k = n - 3$ and $n = k + 3$ into the first series, and $k = n$ into the second series.

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$$6c_3 + \sum_{k=1}^{\infty} c_{k+3}(k+3)(k+2)(k+1)x^k - \sum_{k=1}^{\infty} c_k k x^k = 0$$

With the series in phase and matching indices, we can finally add them.

$$6c_3 + \sum_{k=1}^{\infty} [c_{k+3}(k+1)(k+2)(k+3) - c_k k]x^k = 0$$

We get

$$6c_3 = 0$$

$$c_3 = 0$$

and

$$c_{k+3}(k+1)(k+2)(k+3) - c_k k = 0 \qquad k = 1, 2, 3, \ldots$$

We'll solve the recurrence relation for the coefficient with the largest subscript, c_{k+3}.

$$c_{k+3} = \frac{k}{(k+1)(k+2)(k+3)} c_k \qquad k = 1, 2, 3, \ldots$$

Now we'll start plugging in values $k = 1, 2, 3, \ldots$.

$$k = 1 \quad c_4 = \frac{c_1}{4!} \qquad k = 2 \quad c_5 = \frac{4c_2}{5!} \qquad k = 3 \quad c_6 = 0$$

$$k = 4 \quad c_7 = \frac{4c_1}{7!} \qquad k = 5 \quad c_8 = \frac{20c_2}{8!} \qquad k = 6 \quad c_9 = 0$$

$$\vdots \qquad \vdots \qquad \vdots$$

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Then the general solution is

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + \dots$$

$$y = c_0 + c_1x + c_2x^2 + \frac{1}{4!}c_1x^4 + \frac{4}{5!}c_2x^5 + \frac{4}{7!}c_1x^7 + \frac{20}{8!}c_2x^8 + \dots$$

$$y = c_0 + c_1\left(x + \frac{1}{4!}x^4 + \frac{4}{7!}x^7 + \dots\right) + c_2\left(x^2 + \frac{4}{5!}x^5 + \frac{20}{8!}x^8\right) + \dots$$

■ 2. Find a power series solution in x to the differential equation.

$$y''' - y'' = 0$$

Solution:

Substitute y'' and y''' into the differential equation.

$$\sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-3} - \sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} = 0$$

Now the series are in phase, but the indices don't match. We can substitute $k = n - 3$ and $n = k + 3$ into the first series, and $k = n - 2$ and $n = k + 2$ into the second series.

$$\sum_{k=0}^{\infty} c_{k+3}(k+3)(k+2)(k+1)x^k - \sum_{k=0}^{\infty} c_{k+2}(k+2)(k+1)x^k = 0$$

With the series in phase and matching indices, we can finally add them.

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$$\sum_{k=0}^{\infty}[c_{k+3}(k+3)(k+2)(k+1)-c_{k+2}(k+2)(k+1)]x^k=0$$

We get

$$c_{k+3}(k+3)(k+2)(k+1)-c_{k+2}(k+2)(k+1)=0 \qquad k=0,\ 1,\ 2,\ \ldots$$

We'll solve the recurrence relation for the coefficient with the largest subscript, c_{k+3}.

$$c_{k+3}=\frac{c_{k+2}}{k+3} \qquad k=0,\ 1,\ 2,\ \ldots$$

Now we'll start plugging in values $k=0,\ 1,\ 2,\ldots.$

$$k=0 \qquad c_3=\frac{c_2}{3}$$

$$k=1 \qquad c_4=\frac{c_2}{3\cdot 4}$$

$$k=2 \qquad c_5=\frac{c_2}{3\cdot 4\cdot 5}$$

Then the general solution is

$$y=c_0+c_1x+c_2x^2+c_3x^3+\ldots$$

$$y=c_0+c_1x+c_2x^2+\frac{c_2}{3}x^3+\frac{c_2}{3\cdot 4}x^4+\frac{c_2}{3\cdot 4\cdot 5}x^5+\ldots$$

$$y=c_0+c_1x+c_2x^2+\frac{2c_2}{3!}x^3+\frac{2c_2}{4!}x^4+\frac{2c_2}{5!}x^5+\ldots$$

$$y=c_0+c_1x+c_2\left(x^2+\frac{2}{3!}x^3+\frac{2}{4!}x^4+\frac{2}{5!}x^5+\ldots\right)$$

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■ 3. Find a power series solution in x to the differential equation.

$$xy''' - y' = 0$$

Solution:

Substitute y' and y''' into the differential equation.

$$x\sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-3} - \sum_{n=1}^{\infty} c_n n x^{n-1} = 0$$

$$\sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-2} - \sum_{n=1}^{\infty} c_n n x^{n-1} = 0$$

To put these series in phase, we'll pull the x^0 term out of the second series.

$$\sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-2} - c_1 x^{1-1} - \sum_{n=2}^{\infty} c_n n x^{n-1} = 0$$

$$-c_1 + \sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-2} - \sum_{n=2}^{\infty} c_n n x^{n-1} = 0$$

Now the series are in phase, but the indices don't match. We can substitute $k = n - 2$ and $n = k + 2$ into the first series, and $k = n - 1$ and $n = k + 1$ into the second series.

$$-c_1 + \sum_{k=1}^{\infty} c_{k+2}(k+2)(k+1)k \cdot x^k - \sum_{k=1}^{\infty} c_{k+1}(k+1)x^k = 0$$

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With the series in phase and matching indices, we can finally add them.

$$-c_1 + \sum_{k=1}^{\infty} [c_{k+2}(k+2)(k+1)k - c_{k+1}(k+1)]x^k = 0$$

We get

$$c_1 = 0$$

and

$$c_{k+2}(k+2)(k+1)k - c_{k+1}(k+1) = 0 \qquad k = 1,\ 2,\ 3,...$$

We'll solve the recurrence relation for the coefficient with the largest subscript, c_{k+2}.

$$c_{k+2} = \frac{c_{k+1}}{k(k+2)} \qquad k = 1,\ 2,\ 3,...$$

Now we'll start plugging in values $k = 1,\ 2,\ 3,....$

$$k = 1 \qquad c_3 = \frac{2c_2}{3!}$$

$$k = 2 \qquad c_4 = \frac{c_2}{4!}$$

$$k = 3 \qquad c_5 = \frac{2c_2}{3!5!}$$

$$k = 4 \qquad c_6 = \frac{2c_2}{4!6!}$$

$$k = 5 \qquad c_7 = \frac{2c_2}{5!7!}$$

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$$k = 6 \qquad c_8 = \frac{2c_2}{6!8!}$$

$$\vdots$$

$$c_m = \frac{2c_2}{(m-2)!m!} \qquad m \geq 3$$

Then the general solution is

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$$

$$y = c_0 + c_1 x + c_2 x^2 + \frac{2c_2}{1!3!}x^3 + \frac{2c_2}{2!4!}x^4 + \frac{2c_2}{3!5!}x^5 + \dots$$

$$y = c_0 + c_1 x + \sum_{m=2}^{\infty} \frac{2c_2}{(m-2)!m!}x^m$$

■ 4. Find a power series solution in x to the differential equation.

$$x^2 y''' + y'' - xy = 0$$

Solution:

Substitute y, y'', and y''' into the differential equation.

$$x^2 \sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-3} + \sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} - x\sum_{n=0}^{\infty} c_n x^n = 0$$

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$$\sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-1} + \sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

To put these series in phase, we'll pull the x^0 and x^1 terms out of the second series and pull the x^1 term out of the third series.

$$\sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-1} + c_2 \cdot 2(2-1)x^{2-2} + c_3 \cdot 3(3-1)x^{3-2}$$

$$+ \sum_{n=4}^{\infty} c_n n(n-1)x^{n-2} - c_0 x - \sum_{n=1}^{\infty} c_n x^{x+1} = 0$$

$$2c_2 + (6c_3 - c_0)x + \sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-1} + \sum_{n=4}^{\infty} c_n n(n-1)x^{n-2} - \sum_{n=1}^{\infty} c_n x^{n+1} = 0$$

Now the series are in phase, but the indices don't match. We can substitute $k = n - 3$ and $n = k + 3$ into the first series, $k = n - 4$ and $n = k + 4$ into the second series, and $k = n - 1$ and $n = k + 1$ into the third series.

$$2c_2 + (6c_3 - c_0)x + \sum_{k=0}^{\infty} c_{k+3}(k+3)(k+2)(k+1)x^{k+2}$$

$$+ \sum_{k=0}^{\infty} c_{k+4}(k+4)(k+3)x^{k+2} - \sum_{k=0}^{\infty} c_{k+1}x^{k+2} = 0$$

With the series in phase and matching indices, we can finally add them.

$$2c_2 + (6c_3 - c_0)x$$

$$+ \sum_{k=0}^{\infty} [c_{k+3}(k+1)(k+2)(k+3) + c_{k+4}(k+3)(k+4) - c_{k+1}]x^{k+2} = 0$$

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We get

$$2c_2 = 0$$

$$c_2 = 0$$

and

$$6c_3 - c_0 = 0$$

$$c_3 = \frac{c_0}{6}$$

and

$$c_{k+3}(k+1)(k+2)(k+3) + c_{k+4}(k+3)(k+4) - c_{k+1} = 0 \qquad k = 0,\ 1,\ 2,...$$

We'll solve the recurrence relation for the coefficient with the largest subscript, c_{k+4}.

$$c_{k+4} = \frac{c_{k+1} - c_{k+3}(k+1)(k+2)(k+3)}{(k+3)(k+4)} \qquad k = 0,\ 1,\ 2,...$$

Now we'll start plugging in values $k = 0,\ 1,\ 2,\ \ldots$.

$$k = 0 \qquad c_4 = \frac{c_1 - c_0}{12}$$

$$k = 1 \qquad c_5 = \frac{-c_1 + c_0}{10}$$

$$k = 2 \qquad c_6 = \frac{c_1}{5} - \frac{7}{36}c_0$$

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$$k = 3 \qquad c_7 = \frac{31}{56}c_0 - \frac{41}{72}c_1$$

$$\vdots$$

Then the general solution is

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + \dots$$

$$y = c_0 + c_1x + \frac{c_0}{6}x^3 + \frac{c_1 - c_0}{12}x^4 + \frac{-c_1 + c_0}{10}x^5$$

$$+\left(\frac{c_1}{5} - \frac{7}{36}c_0\right)x^6 + \left(\frac{31}{56}c_0 - \frac{41}{72}c_1\right)x^7 + \dots$$

$$y = c_0\left(1 + \frac{1}{6}x^3 - \frac{1}{12}x^4 + \frac{1}{10}x^5 - \frac{7}{36}x^6 + \frac{31}{56}x^7 - \dots\right)$$

$$+c_1\left(x + \frac{1}{12}x^4 - \frac{1}{10}x^5 + \frac{1}{5}x^6 - \frac{41}{72}x^7 + \dots\right)$$

■ 5. Find a power series solution in x to the differential equation.

$$y''' + xy' + y = 0$$

Solution:

Substitute y, y', and y''' into the differential equation.

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$$\sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-3} + x\sum_{n=1}^{\infty} c_n n x^{n-1} + \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-3} + \sum_{n=1}^{\infty} c_n n x^{n} + \sum_{n=0}^{\infty} c_n x^n = 0$$

To put these series in phase, we'll pull the x^0 term out of the first and third series.

$$c_3 \cdot 3(3-1)(3-2)x^{3-3} + \sum_{n=4}^{\infty} c_n n(n-1)(n-2)x^{n-3} + \sum_{n=1}^{\infty} c_n n x^n$$

$$+c_0 x^0 + \sum_{n=1}^{\infty} c_n x^n = 0$$

$$6c_3 + c_0 + \sum_{n=4}^{\infty} c_n n(n-1)(n-2)x^{n-3} + \sum_{n=1}^{\infty} c_n n x^n + \sum_{n=1}^{\infty} c_n x^n = 0$$

Now the series are in phase, but the indices don't match. We can substitute $k = n - 3$ and $n = k + 3$ into the first series, and $k = n$ into the second and third series.

$$6c_3 + c_0 + \sum_{k=1}^{\infty} c_{k+3}(k+3)(k+2)(k+1)x^k + \sum_{k=1}^{\infty} c_k k x^k + \sum_{k=1}^{\infty} c_k x^k = 0$$

With the series in phase and matching indices, we can finally add them.

$$6c_3 + c_0 + \sum_{k=1}^{\infty} [c_{k+3}(k+3)(k+2)(k+1) + c_k(k+1)]x^k = 0$$

We get

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$$6c_3 + c_0 = 0$$

$$c_3 = -\frac{c_0}{6}$$

and

$$c_{k+3}(k+3)(k+2)(k+1) + c_k(k+1) = 0 \qquad k = 1, 2, 3...$$

We'll solve the recurrence relation for the coefficient with the largest subscript, c_{k+3}.

$$c_{k+3} = -\frac{c_k}{(k+3)(k+2)} \qquad k = 1, 2, 3...$$

Now we'll start plugging in values $k = 1, 2, 3....$

$$k = 0 \quad c_3 = -\frac{c_0}{6} \qquad k = 1 \quad c_4 = -\frac{c_1}{3 \cdot 4} \qquad k = 2 \quad c_5 = -\frac{c_2}{4 \cdot 5}$$

$$k = 3 \quad c_6 = \frac{c_0}{2 \cdot 3 \cdot 5 \cdot 6} \qquad k = 4 \quad c_7 = \frac{c_1}{3 \cdot 4 \cdot 6 \cdot 7} \qquad k = 5 \quad c_8 = \frac{c_2}{4 \cdot 5 \cdot 7 \cdot 8}$$

$$\vdots \qquad \vdots \qquad \vdots$$

Then the general solution is

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + \dots$$

$$y = c_0 + c_1x + c_2x^2 - \frac{c_0}{6}x^3 - \frac{c_1}{3 \cdot 4}x^4 - \frac{c_2}{4 \cdot 5}x^5 + \frac{c_0}{2 \cdot 3 \cdot 5 \cdot 6}x^6 + \frac{c_1}{3 \cdot 4 \cdot 6 \cdot 7}x^7 + \dots$$

$$y = c_0\left(1 - \frac{1}{6}x^3 + \frac{c_0}{2 \cdot 3 \cdot 5 \cdot 6}x^6 - \dots\right) + c_1\left(x - \frac{c_1}{3 \cdot 4}x^4 + \frac{c_1}{3 \cdot 4 \cdot 6 \cdot 7}x^7 - \dots\right)$$

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$$+c_2\left(x^2-\frac{c_2}{4\cdot 5}x^5+\ldots\right)$$

■ 6. Find a power series solution in x to the differential equation.

$$y'''+xy''-y=0$$

Solution:

Substitute y, y'', and y''' into the differential equation.

$$\sum_{n=3}^{\infty}c_n n(n-1)(n-2)x^{n-3}+x\sum_{n=2}^{\infty}c_n n(n-1)x^{n-2}-\sum_{n=0}^{\infty}c_n x^n=0$$

$$\sum_{n=3}^{\infty}c_n n(n-1)(n-2)x^{n-3}+\sum_{n=2}^{\infty}c_n n(n-1)x^{n-1}-\sum_{n=0}^{\infty}c_n x^n=0$$

To put these series in phase, we'll pull the x^0 terms out of the first and third series.

$$c_3\cdot 3(3-1)(3-2)x^{3-3}+\sum_{n=4}^{\infty}c_n n(n-1)(n-2)x^{n-3}+\sum_{n=2}^{\infty}c_n n(n-1)x^{n-1}$$

$$-c_0x^0-\sum_{n=1}^{\infty}c_n x^n=0$$

$$6c_3-c_0+\sum_{n=4}^{\infty}c_n n(n-1)(n-2)x^{n-3}+\sum_{n=2}^{\infty}c_n n(n-1)x^{n-1}-\sum_{n=1}^{\infty}c_n x^n=0$$

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Now the series are in phase, but the indices don't match. We can substitute $k = n - 3$ and $n = k + 3$ into the first series, $k = n - 1$ and $n = k + 1$ into the second series, and $k = n$ into the third series.

$$6c_3 - c_0 + \sum_{k=1}^{\infty} c_{k+3}(k+3)(k+2)(k+1)x^k + \sum_{k=1}^{\infty} c_{k+1}(k+1)kx^k - \sum_{k=1}^{\infty} c_k x^k = 0$$

With the series in phase and matching indices, we can finally add them.

$$6c_3 - c_0 + \sum_{k=1}^{\infty} [c_{k+3}(k+3)(k+2)(k+1) + c_{k+1}(k+1)k - c_k]x^k = 0$$

We get

$$6c_3 - c_0 = 0$$

$$c_3 = \frac{c_0}{6}$$

and

$$c_{k+3}(k+3)(k+2)(k+1) + c_{k+1}(k+1)k - c_k = 0 \qquad k = 1, 2, 3...$$

We'll solve the recurrence relation for the coefficient with the largest subscript, c_{k+3}.

$$c_{k+3} = \frac{c_k - c_{k+1}(k+1)k}{(k+3)(k+2)(k+1)} \qquad k = 1, 2, 3...$$

Now we'll start plugging in values $k = 1, 2, 3....$

$$k = 1 \qquad c_4 = \frac{c_1 - 2c_2}{24}$$

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$$k = 2 \qquad c_5 = \frac{c_2 - c_0}{60}$$

$$k = 3 \qquad c_6 = \frac{c_0 - 3c_1 + 6c_2}{720}$$

$$k = 4 \qquad c_7 = \frac{8c_0 + c_1 - 10c_2}{5{,}040}$$

$$\vdots$$

Then the general solution is

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + c_6 x^6 + \dots$$

$$y = c_0 + c_1 x + c_2 x^2 + \frac{c_0}{6} x^3 + \frac{c_1 - 2c_2}{24} x^4 + \frac{c_2 - c_0}{60} x^5 + \frac{c_0 - 3c_1 + 6c_2}{720} x^6 + \frac{8c_0 + c_1 - 10c_2}{5{,}040} x^7 + \dots$$

$$y = c_0 \left(1 + \frac{1}{6} x^3 - \frac{1}{60} x^5 + \frac{1}{720} x^6 + \frac{1}{630} x^7 + \dots\right) + c_1 \left(x + \frac{1}{24} x^4 - \frac{1}{240} x^6 + \frac{1}{5{,}040} x^7 + \dots\right) + c_2 \left(x^2 - \frac{1}{12} x^4 + \frac{1}{60} x^5 + \frac{1}{120} x^6 - \frac{1}{504} x^7 + \dots\right)$$

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